

CST8911 Midterm Project

Course: CST8911 Introduction to Cloud Computing

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[GitHub Link: https://github.com/bosichen-ac/CST8911-Midterm](https://github.com/bosichen-ac/CST8911-Midterm)

Scenario Overview

In this project, an Azure Virtual Machine using the most cost-effective instance is created to act as a secure management station and testing hub.

A Cosmos DB is created and an Azure Function App is used to interact with the database using RESTful API functions.

The Azure Key Vault is used for storing the access keys to the Cosmos DB and Function App.

Security Architecture using Key Vault

When trying to call the Function App

1. The Function App authenticates Key Vault using its System-Assigned Managed Identity.
2. Azure verifies the identity.
3. The Function App requests access to secrets in Key Vault.
4. Key Vault checks permissions using Role-Based Access Control (RBAC).
5. The secret values are sent back to the Function App for further actions.

Workflow

1. Create Resource Group

The screenshot shows the Microsoft Azure portal interface for creating a new resource group. The browser address bar indicates the URL is `portal.azure.com/#view/HubsExtension/ResourceGroupCreate.ReactView/_provisioningContext~/%7B%22telemetryId%22%3A%22...`. The page title is "Create a resource group". The "Basics" tab is active, showing a description of a resource group and three input fields: "Subscription" (set to "Azure for Students"), "Resource group name" (set to "midterm"), and "Region" (set to "(Canada) Canada Central"). At the bottom, there are three buttons: "Previous", "Next", and "Review + create" (which is highlighted in blue).

2. Create a Virtual Machine and Save the SSH key

Create a virtual machine

Save the key pair file in `~/.ssh` and use `chmod 400 cst8911-vm_key.pem` to secure the key. This removes all access permissions so that only the owner can read it.

Create a virtual machine - Microsoft Azure

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a low cost VMHelp me choose the right VM size for my workload+1

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * midterm
[Create new](#)

Instance details

Virtual machine name * cst8911-vm

Region * (Canada) Canada Central
[Deploy to an Azure Extended Zone](#)

Availability options Availability zone

Zone options
☐ Self-selected zone
Choose up to 3 availability zones, one VM per zone
☒ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

< PreviousNext : Disks >Review + create

[Give feedback](#)

Create a virtual machine - Microsoft Azure

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

Microsoft Azure

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Create a virtual machine

Help me create a low cost VMHelp me choose the right VM size for my workload+1

Availability zone ⓘ

Azure-selected zone (Preview)

Security type ⓘ

Trusted launch virtual machines

Configure security features

Image * ⓘ

Ubuntu Server 24.04 LTS - x64 Gen2

See all images | Configure VM generation

VM architecture ⓘ

Arm64

x64

Run with Azure Spot discount ⓘ

☐

Size * ⓘ

Standard_B2ats_v2 - 2 vcpus, 1 GiB memory (US\$7.67) (free services eligible)

See all sizes

Enable Hibernation ⓘ

☐

ⓘ

Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. [Learn more](#)

Administrator account

Authentication type ⓘ

SSH public key

Password

ⓘ

Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username * ⓘ

azureuser

< Previous

Next : Disks >

Review + create

Give feedback

portal.azure.com/#create/Microsoft.VirtualMachine-ARM

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Create a virtual machine

Help me create a low cost VM Help me choose the right VM size for my workload +1

Authentication type

☒ SSH public key

☐ Password

Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username *

azureuser

SSH public key source

Generate new key pair

SSH Key Type

☒ RSA SSH Format

☐ Ed25519 SSH Format

Ed25519 provides a fixed security level of no more than 128 bits for 256-bit key, while RSA could offer better security with keys longer than 3072 bits.

Key pair name *

cst8911-vm_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *

☐ None

☒ Allow selected ports

Select inbound ports *

SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

< Previous Next : Disks > Review + create Give feedback

The B2ats (2vcpus, 1GB memory) provides the lowest cost of computing for Canada Central location and is enough for development and testing.

3. Create Cosmos DB (NoSQL option)

Create account

Create a **serverless** account for NoSQL named **cosmos-midterm**.

Create Azure Cosmos DB Account

portal.azure.com/#create/Microsoft.DocumentDB

Microsoft Azure

Search resources, services, and docs (G+)

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Home

Azure Cosmos DB

Create an Azure Cosmos DB account

Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Basics

Global distribution

Networking

Backup Policy

Security

Tags

Review + create

Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. Go to production starting at \$24/month per database, multiple containers included. [Learn more](#)

Project Details

Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.

Workload Type * ⓘ

Learning

Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource Group *

midterm

[Create new](#)

Instance Details

Account Name *

cosmos-midterm

Configure availability zone settings for your account. You cannot change these settings once the account is created.

Availability Zones ⓘ

☐ Enable ☒ Disable

Location * ⓘ

(Canada) Canada Central

Available locations are determined by your subscription's access and availability zone support (if that is enabled). If you don't see or cannot select your desired location, please open a support request for region access. [Click here for more details on how to create a region access request](#)

Review + create

Previous

Next: Global distribution

[Feedback](#)

portal.azure.com/#create/Microsoft.DocumentDB

Microsoft Azure Search resources, services, and docs (G+/) Copilot chen0854@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > Azure Cosmos DB > Create an Azure Cosmos DB account

Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource Group * midterm [Create new](#)

Instance Details

Account Name * cosmos-midterm

Configure availability zone settings for your account. You cannot change these settings once the account is created.

Availability Zones ☐ Enable ☒ Disable

Location * (Canada) Canada Central

Available locations are determined by your subscription's access and availability zone support (if that is enabled). If you don't see or cannot select your desired location, please open a support request for region access. [Click here for more details on how to create a region access request](#)

Capacity mode

☐ Provisioned throughput
Request units per second (RU/s) are pre-configured and billed hourly, offering guaranteed throughput with two modes— **Autoscale**, which dynamically adjusts within a 1-10x range for varying workloads, and **Manual**, with fixed RU/s for steady, predictable traffic.

☒ **Serverless**
Ideal for intermittent or unpredictable traffic. Billed only for consumed RU/s, scaling on-demand without capacity planning or minimum charges. [Learn more about capacity mode](#)

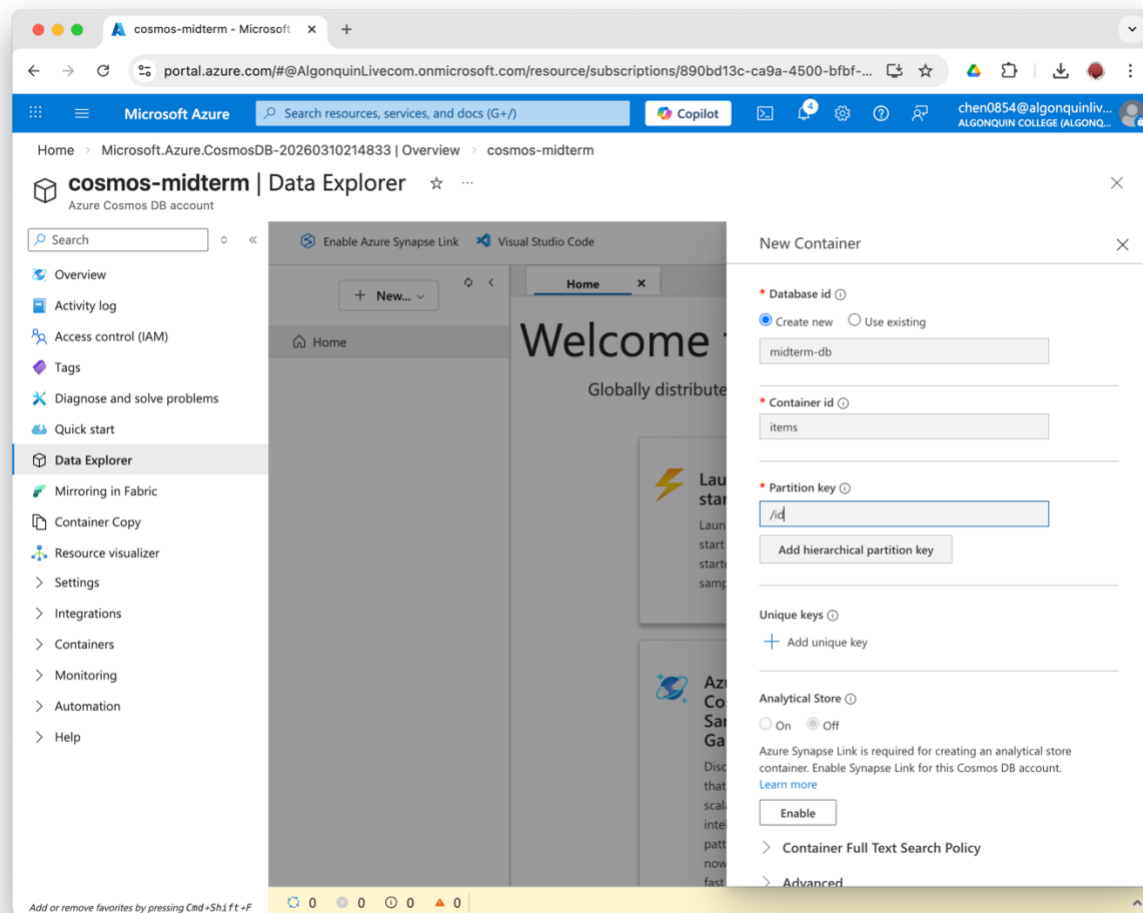
[Review + create](#) [Previous](#) [Next: Global distribution](#) [Feedback](#)

Serverless mode is selected because it charges only for request units consumed, and is the ideal for development and testing purposes.

Create database

Create database **midterm-db**.

Create a container **items** within the database, with **/id** as the partition key.



4. Create Function App

Create the Function App under Consumption Plan, named **function-midterm**.

Runtime stack: Node.js, 22LTS

Create Function App (Consumption)

portal.azure.com/#view/WebsitesExtension/AppServiceFunctionAppCreateV3Blade/_provisioningContext~/%7...

Microsoft Azure

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Home > Create Function App

Create Function App (Consumption (Windows))

publish a web application.

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource Group *

midterm

Create new

Instance Details

Function App name *

function-midterm

-gqehg5duhzdtfuaf.canadacentral-01.azurewebsites.net

Secure unique default hostname on. [More about this update](#)

Operating System

Windows

For Linux, [learn more](#)

Flex Consumption is now the recommended serverless hosting plan for Azure Functions. [Learn more.](#)

Runtime stack *

Node.js

Version *

22 LTS

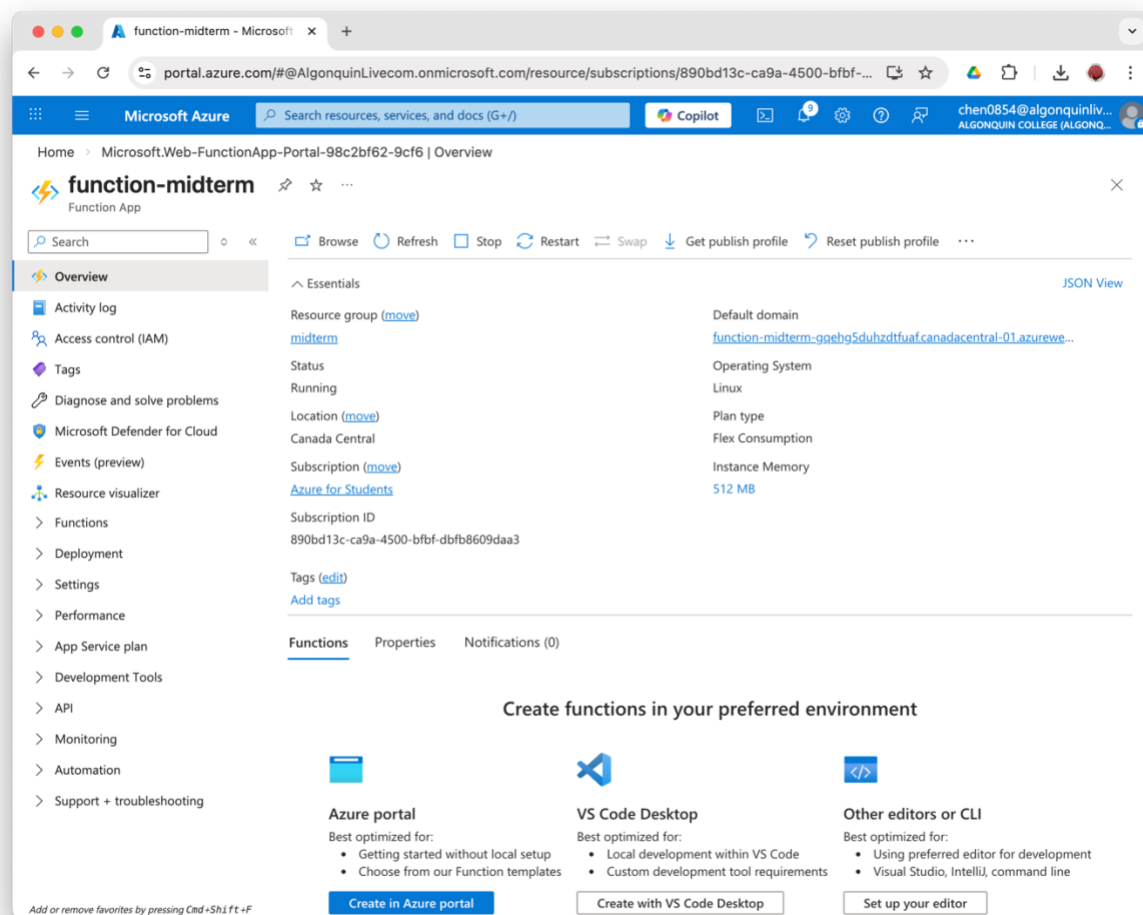
Region *

Canada Central

Review + create

< Previous

Next : Storage >



The Consumption plan is a serverless option and charges only when the function is executing. It is the most cost-effective choice for event-driven but low-traffic functions.

5. Create Key Vault

function-midterm - Microsoft x | cosmos-midterm - Microsoft x | Create a key vault - Microsoft x +

portal.azure.com/#view/Microsoft_Azure_KeyVault/CreateKeyVaultBlade/_provisioningContext~/%7B"initialVal...

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Home > Key vaults

Create a key vault

Basics Access configuration Networking Tags Review + create

Azure Key Vault is a cloud service used to manage keys, secrets, and certificates. Key Vault eliminates the need for developers to store security information in their code. It allows you to centralize the storage of your application secrets which greatly reduces the chances that secrets may be leaked. Key Vault also allows you to securely store secrets and keys backed by Hardware Security Modules or HSMs. The HSMs used are Federal Information Processing Standards (FIPS) 140-2 Level 2 validated. In addition, key vault provides logs of all access and usage attempts of your secrets so you have a complete audit trail for compliance.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * midterm [Create new](#)

Instance details

Key vault name * kv-cst8911midterm ✓

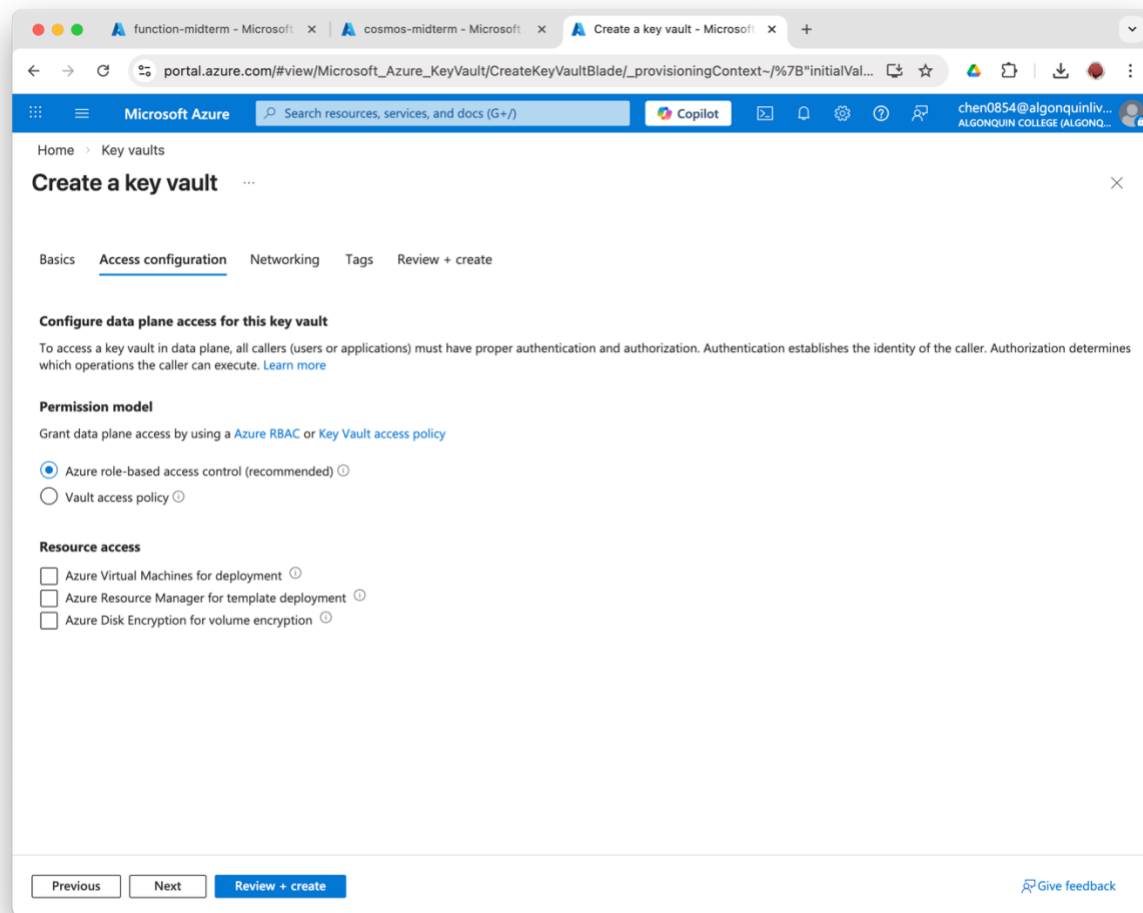
Region * Canada Central

Pricing tier * Standard

Recovery options

Soft delete protection will automatically be enabled on this key vault. This feature allows you to recover or permanently delete a key vault and secrets for the duration of the retention period. This protection applies to the key vault and the secrets stored within the key vault.

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)



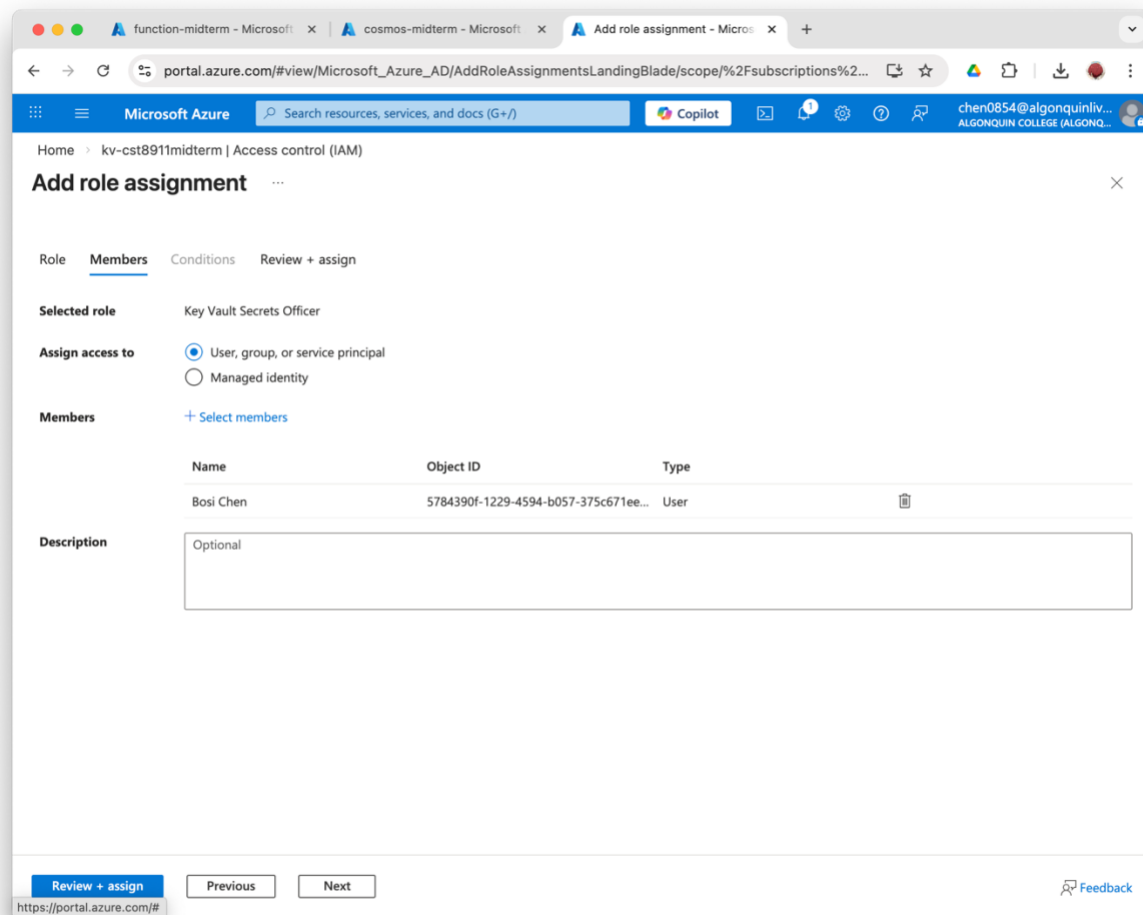
Selecting the Azure role-based access control can grant access by using the RBAC.

Azure Key Vault is used to store sensitive configuration values such as API access keys. It prevents secrets from being stored directly in code or configuration files.

The standard pricing tier is the most cost-effective choice for the Key Vault.

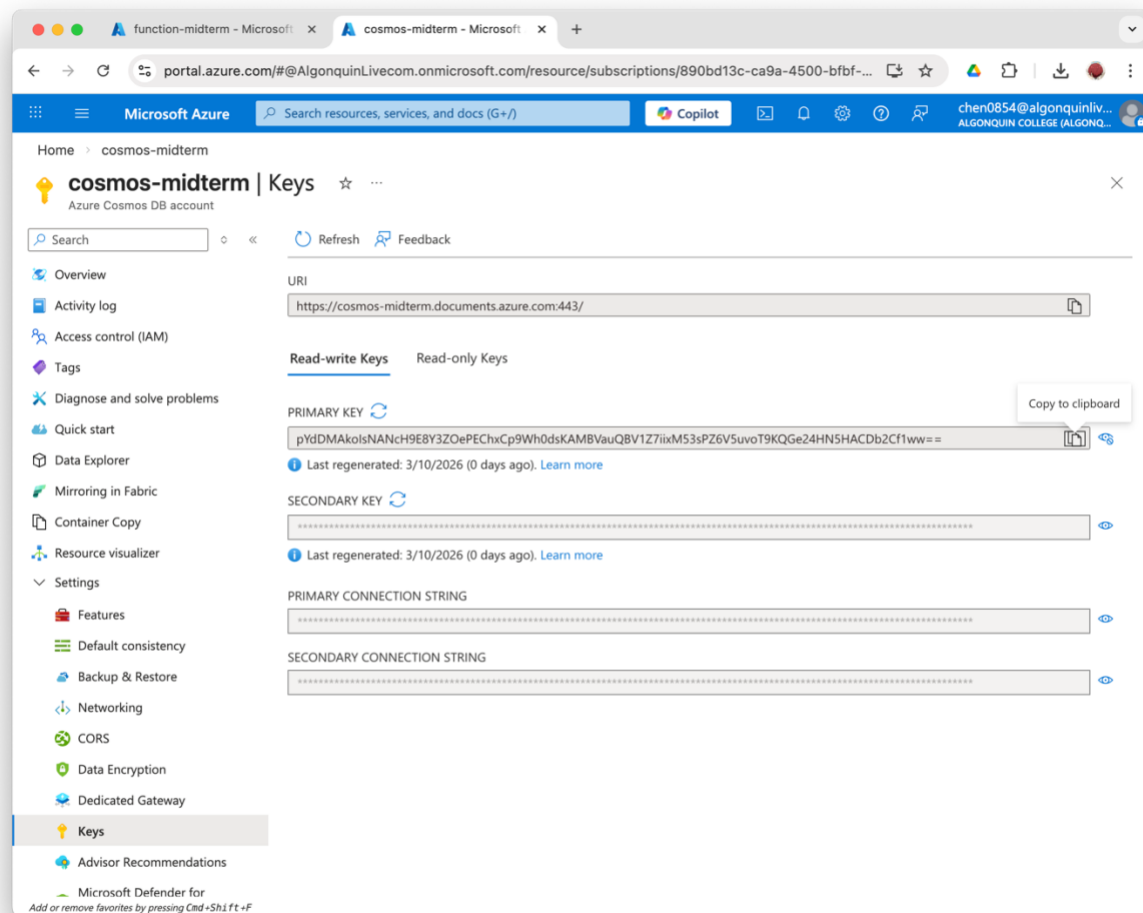
6. Store Keys in Key Vault

First, grant the user the Key Vault Secret Officer Role in the Access control(IAM) of the Key Vault.



Store Cosmos DB Primary Key in Key Vault.

Generate a Secret in Key Vault named **cosmos-key** with the Cosmos DB key as value.



function-midterm - Microsoft A x cosmos-midterm - Microsoft A x Create a secret - Microsoft A x +

portal.azure.com/#view/Microsoft_Azure_KeyVault/CreateSecretBlade/secret~/null/vaultId/%2Fsubscriptions/... Copilot chen0854@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > kv-cst8911midterm | Secrets

Create a secret

Upload options: Manual

Name *: cosmos-key ✓

Secret value *: ***** ✓

Content type (optional):

Set activation date: ☐

Set expiration date: ☐

Enabled: Yes No

Tags: 0 tags

Create Cancel

Add function access key to Key Vault.

Generate a Secret in Key Vault named **function-default-key** with the default function app key as value.

function-midterm - Microsoft

cosmos-midterm - Microsoft

df021fc966a14fc49cb6e25b -

portal.azure.com/#@AlgonquinLivecom.onmicrosoft.com/resource/subscriptions/890bd13c-ca9a-4500-...

Microsoft Azure

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Home > Microsoft.Web-FunctionApp-Portal-98c2bf62-9cf6 | Overview > function-midterm

function-midterm | App keys

Function App

Search

+ New host key Refresh Send us your feedback

Microsoft Defender for Cloud

Events (preview)

Resource visualizer

Functions

App keys

App files

Proxies

Deployment

Settings

Environment variables

Configuration

Instances

Durable Functions

Authentication

Azure OpenAI

Vector Database

Identity

Backups

Custom domains

Certificates

System keys

System keys are automatically managed by the Function runtime. System Keys provide granular access to functions runtime features.

Name	Value
------	-------

Host keys (all functions)

Use Host keys with your clients to access all your HTTP functions in the app. _master key grants admin access to Functions Runtime APIs.

Name	Value		
_master		Copied Show value	Renew key value
default		Show value	Renew key value

Add or remove favorites by pressing Cmd+Shift+F

function-midterm - Microsoft A...cosmos-midterm - Microsoft A...Create a secret - Microsoft A...+

portal.azure.com/#view/Microsoft_Azure_KeyVault/CreateSecretBlade/secret~/null/vaultId/%2Fsubscripti...

Microsoft AzureSearch resources, services, and docs (G+/)Copilot

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Home > kv-cst8911midterm | Secrets

Create a secret

Upload options

Manual

Name *

function-default-key

Secret value *

Content type (optional)

Set activation date

☐

Set expiration date

☐

Enabled

YesNo

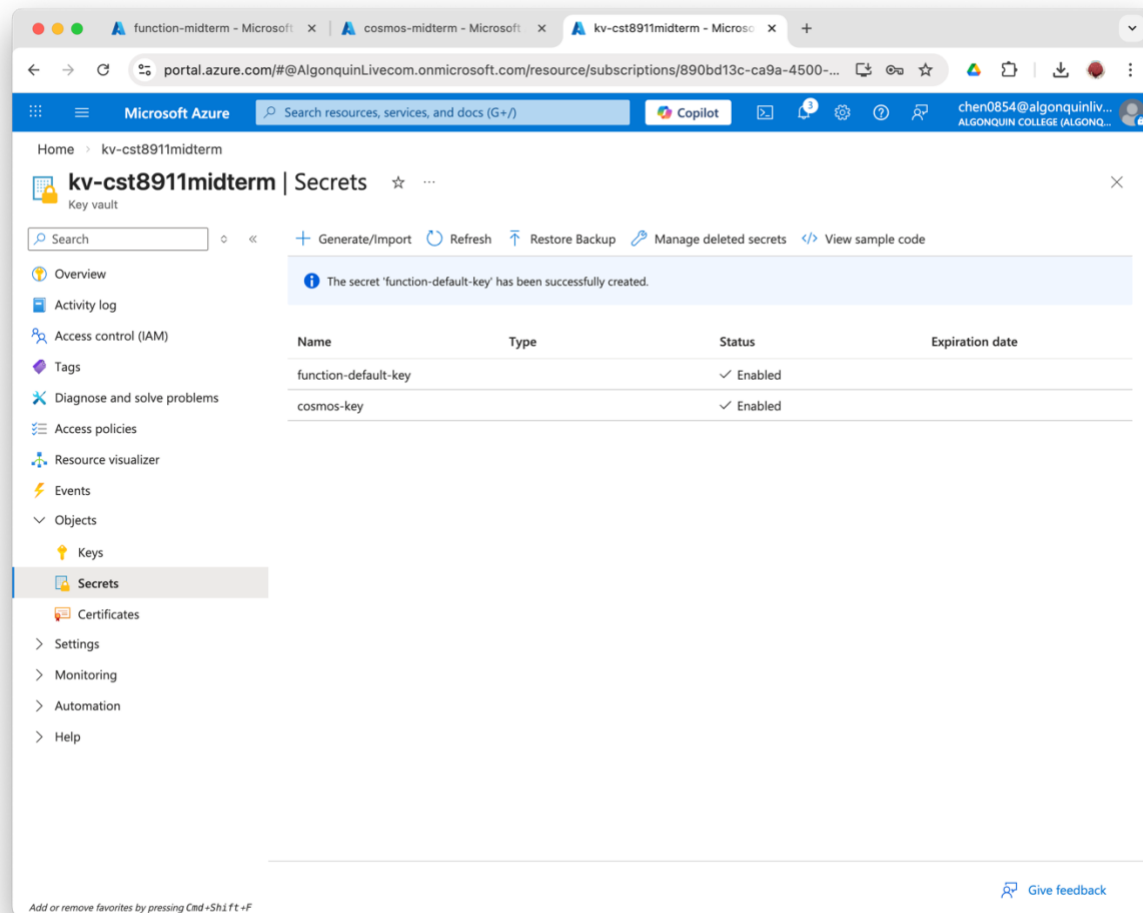
Tags

0 tags

Create

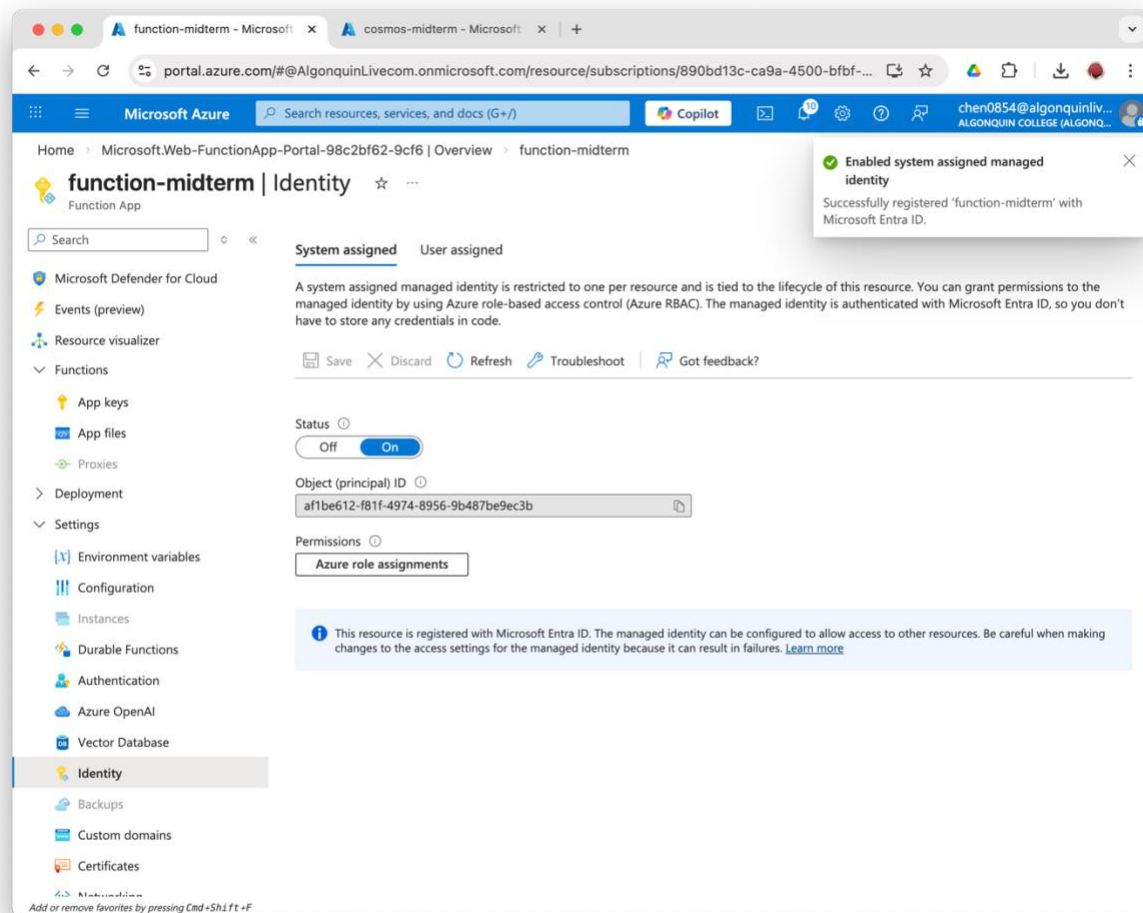
Cancel

Final Overview

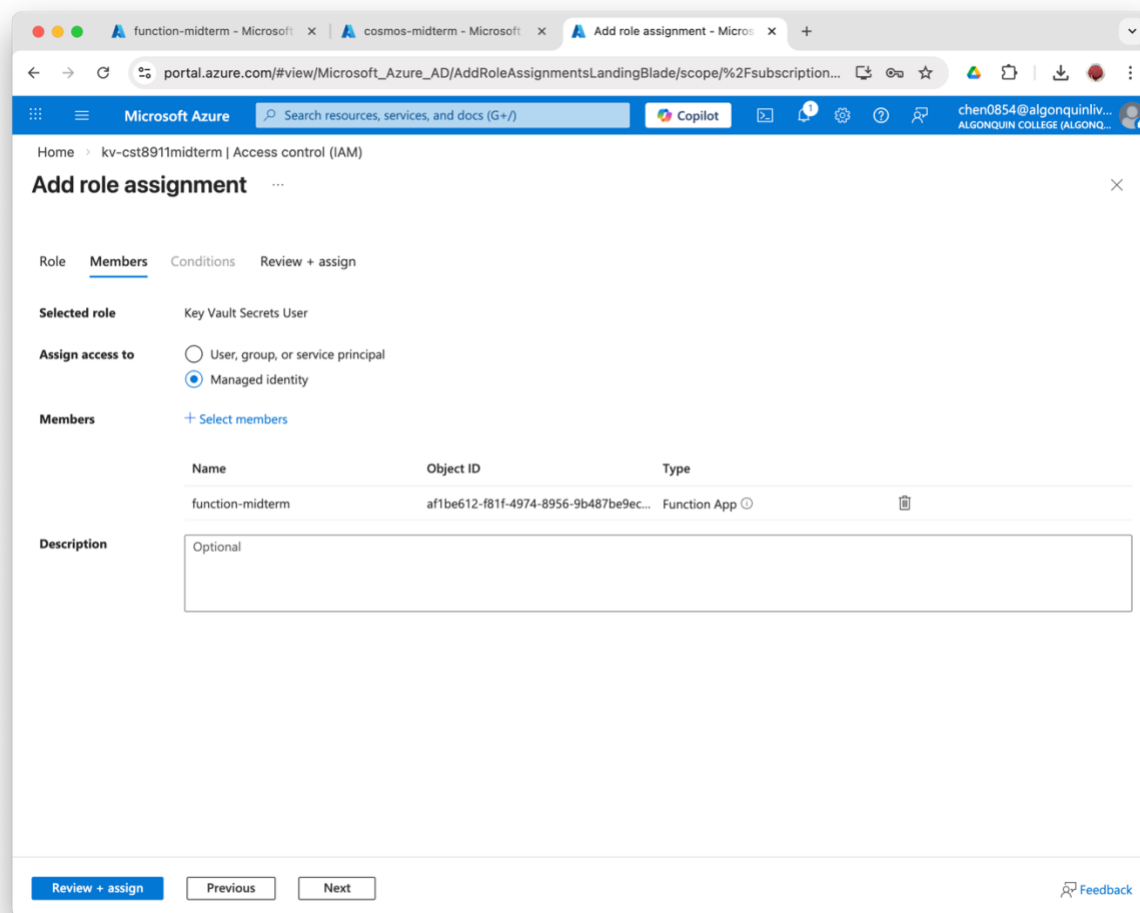


7. Give Function App access to only this secret

Turn on the System-Assigned Managed Identity.



Then, in the Access Control IAM, assign Key Vault Secrets User role to the Function App's Managed Identity.



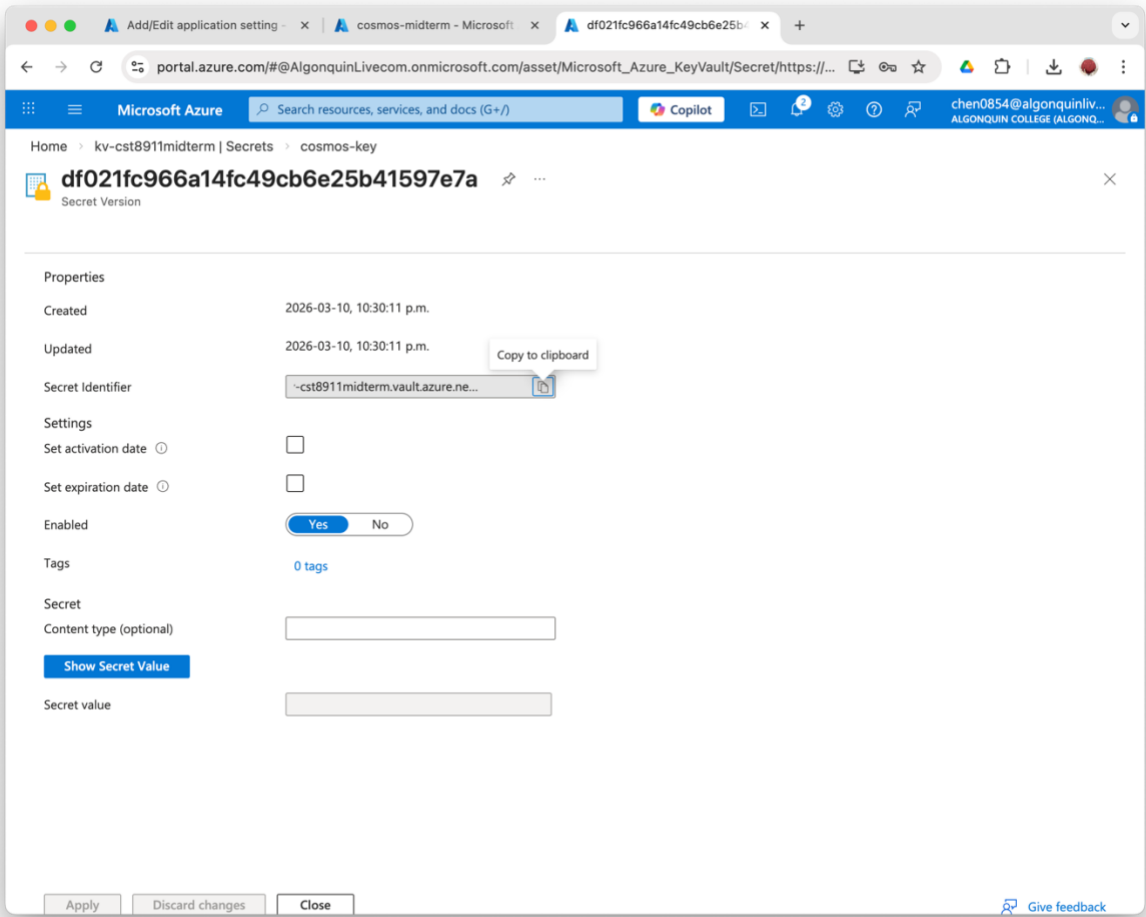
The Function App uses the System-Assigned Managed Identity to allow it to be authorized by Azure-managed services. It improves security by eliminating the direct storing of secret values in application services and thus reduces the risk of leaks.

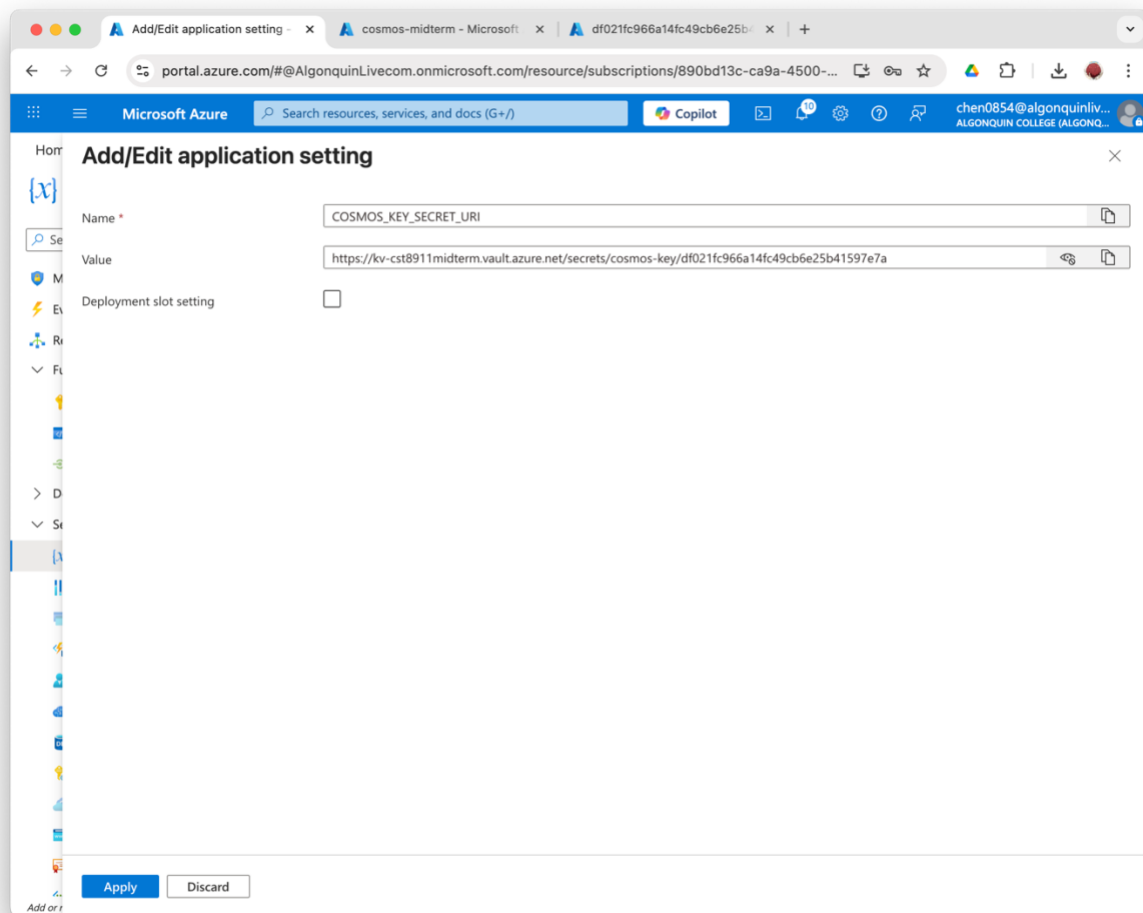
The Function App's System-Assigned Managed Identity is granted the Key Vault Secrets User role, allowing it to only read and write data according to the least-privilege principle.

8. Set Up the Function App Environments

Name	Value
COSMOS_KEY_SECRET_URI	Secret Identifier of cosmos-key

COSMOS_ENDPOINT	Cosmos DB URI (from Overview Page)
COSMOS_DB_NAME	midterm-db
COSMOS_CONTAINER_NAME	items





function-midterm - Microsoft | cosmos-midterm - Microsoft | df021fc966a14fc49cb6e25b | +

portal.azure.com/#@AlgonquinLivecom.onmicrosoft.com/resource/subscriptions/890bd13c-ca9a-4500-...

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function-midterm | Environment variables

Function App

Search

Microsoft Defender for Cloud

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Authentication

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App settings

Connection strings

Search

+ Add

Refresh

Show values

Advanced edit

Pull reference values

Name	Value	Deployment slot setting	Source	Delete
APPLICATIONINSIGHTS_CON...	Show value		App Service	
AzureWebJobsStorage	Show value		App Service	
COSMOS_KEY_SECRET_URI	Show value		App Service	
FUNCTIONS_EXTENSION_VER...	Show value		App Service	
FUNCTIONS_WORKER_RUNTI...	Show value		App Service	
WEBSITE_CONTENTAZUREFILE...	Show value		App Service	
WEBSITE_CONTENTSHARE	Show value		App Service	
WEBSITE_NODE_DEFAULT_VER...	Show value		App Service	

Apply

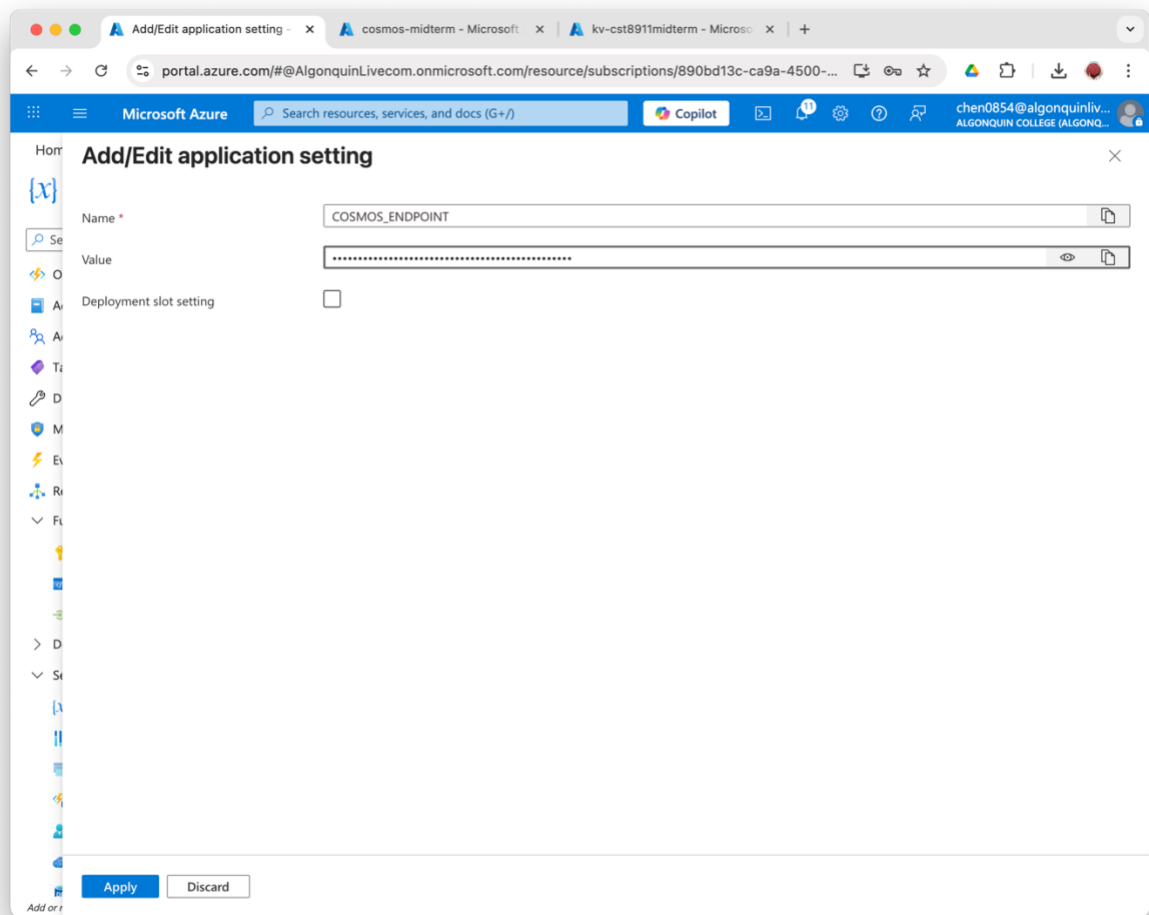
Discard

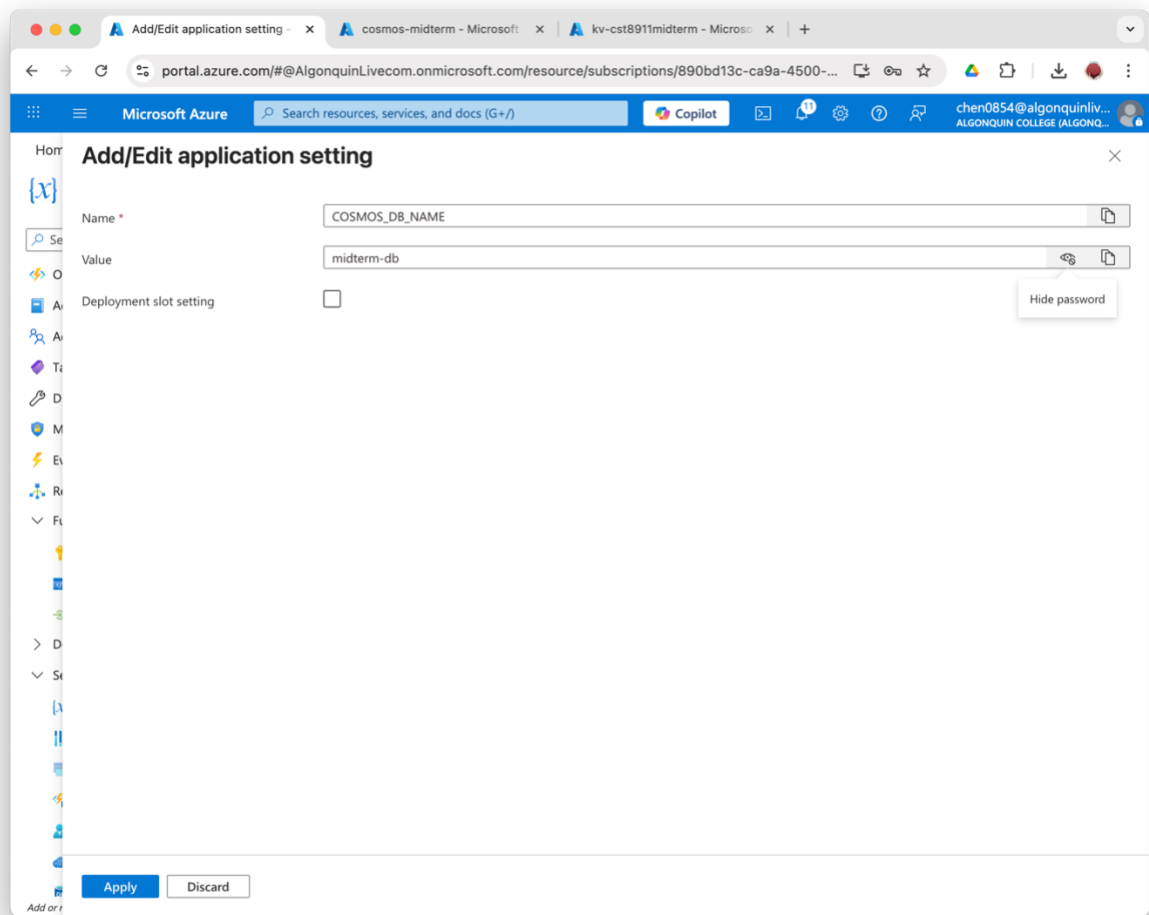
Send us your feedback

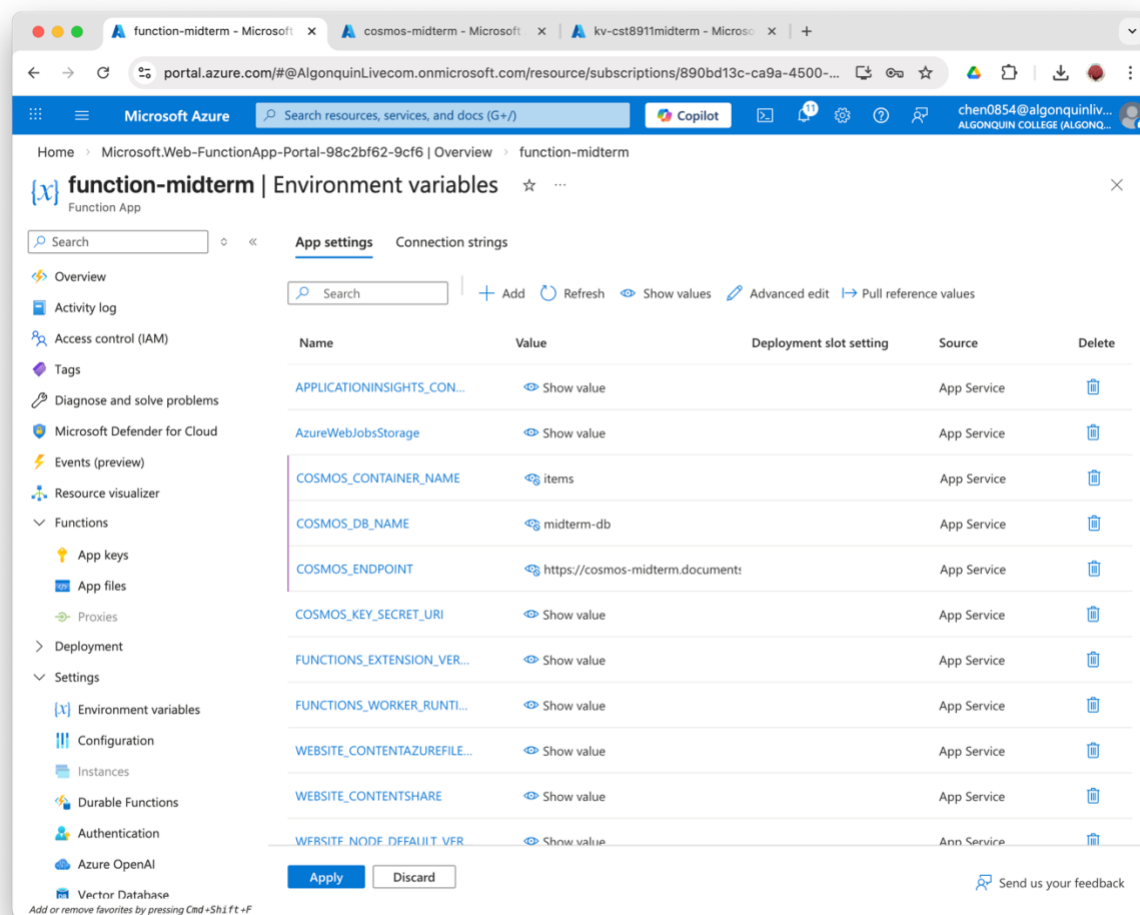
Updating app settings

Successfully updated app settings

Add or remove favorites by pressing Cmd+Shift+F







Store all the secret configuration strings in the App Environments to make sure that all the sensitive configuration data is kept secret separately instead of being hard-coded.

9. Create the functions in the Function App and Test/Run

Use Microsoft Copilot to generate 3 HTTP-triggered Functions, with Authorization level

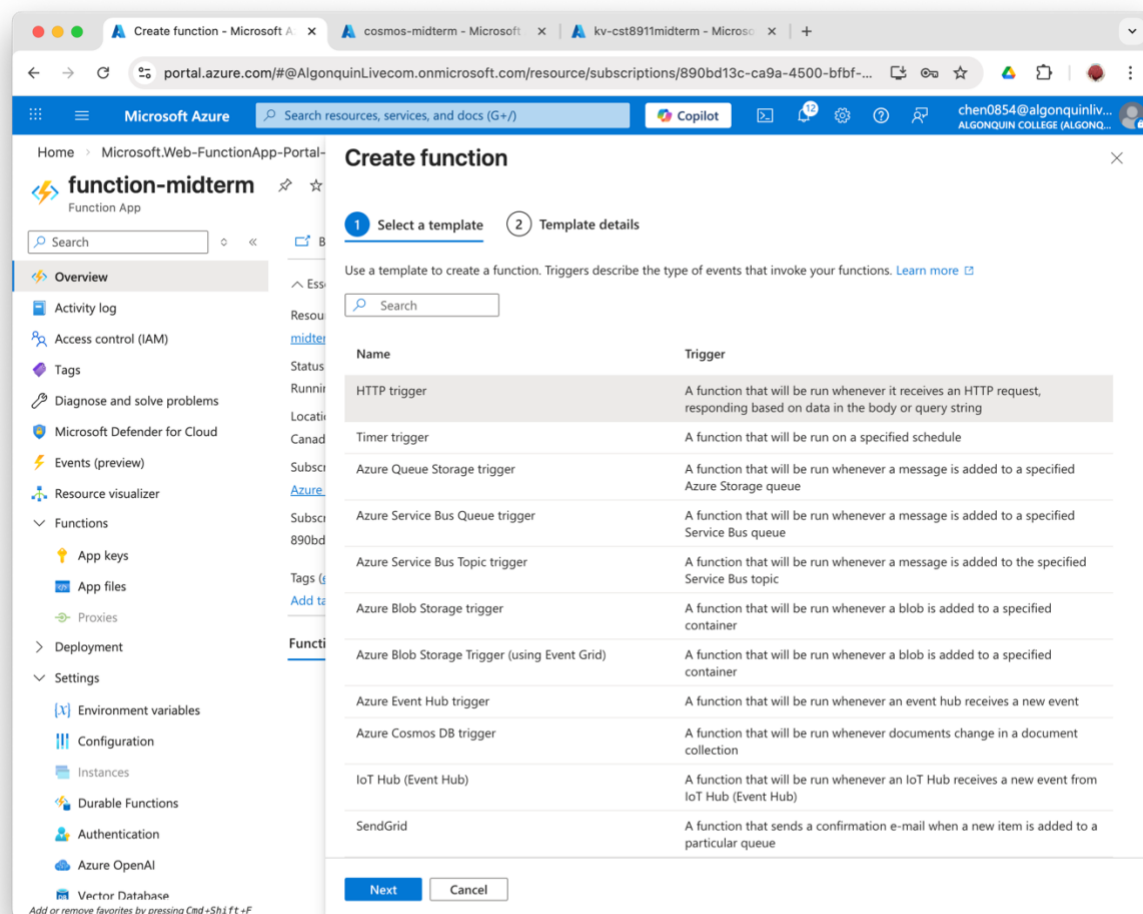
Function:

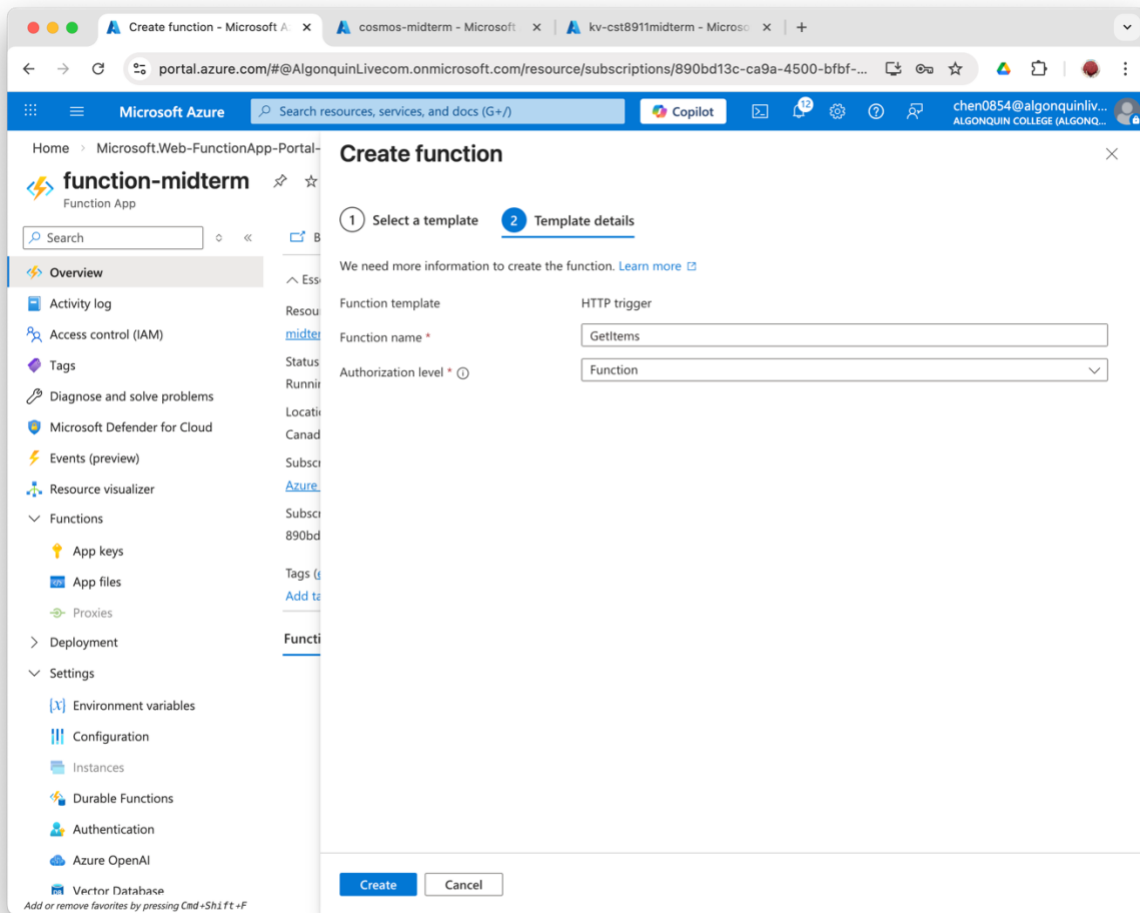
Function name	API end	What it does
GetItems (GET)	GET /items	Get all items from midterm-db

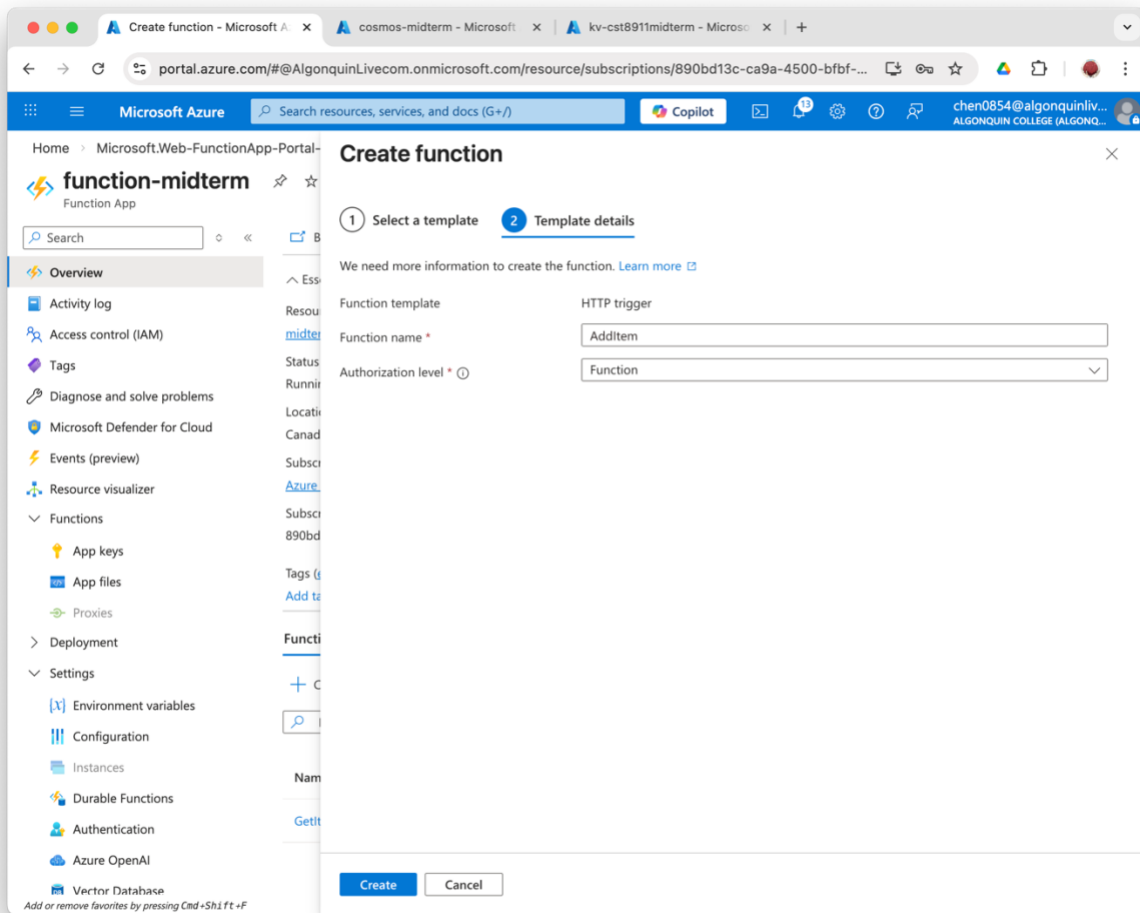
AddItem (POST)	POST /items	Insert a new item to midterm-db
DeleteItem (DELETE)	DELETE /items/{id}	Remove an item by ID

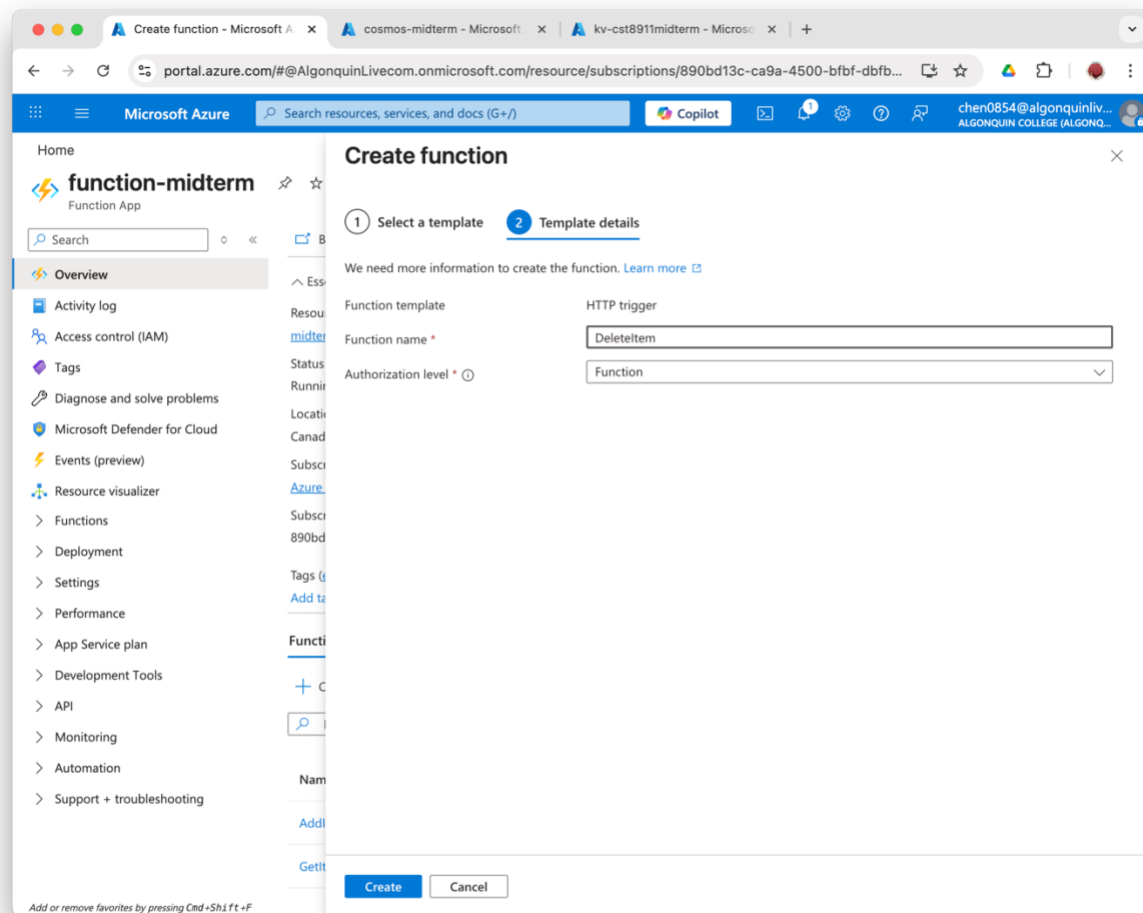
The API communicates with Cosmos DB using Node.js. An API key is required for accessing the Authorization level Function to ensure that only requests with a valid function access key can touch the API endpoints.

Create the three functions









Add dependencies

Go to Development Tools --> Advanced Tools --> Go --> Debug Console --> PowerShell, create a **package.json** file.

function-midterm - Microsoft x Diagnostic Console x cosmos-midterm - Microsoft x kv-cst8911midterm - Microsoft x +

function-midterm-gqehg5duhzdtfuaf.scm.canadacentral-01.azurewebsites.net/DebugConsole/?shell=powershell

Kudu+ Environment Debug console Process explorer Tools Site extensions chen0854@algonquinlive.com

... / wwwroot + 6 items

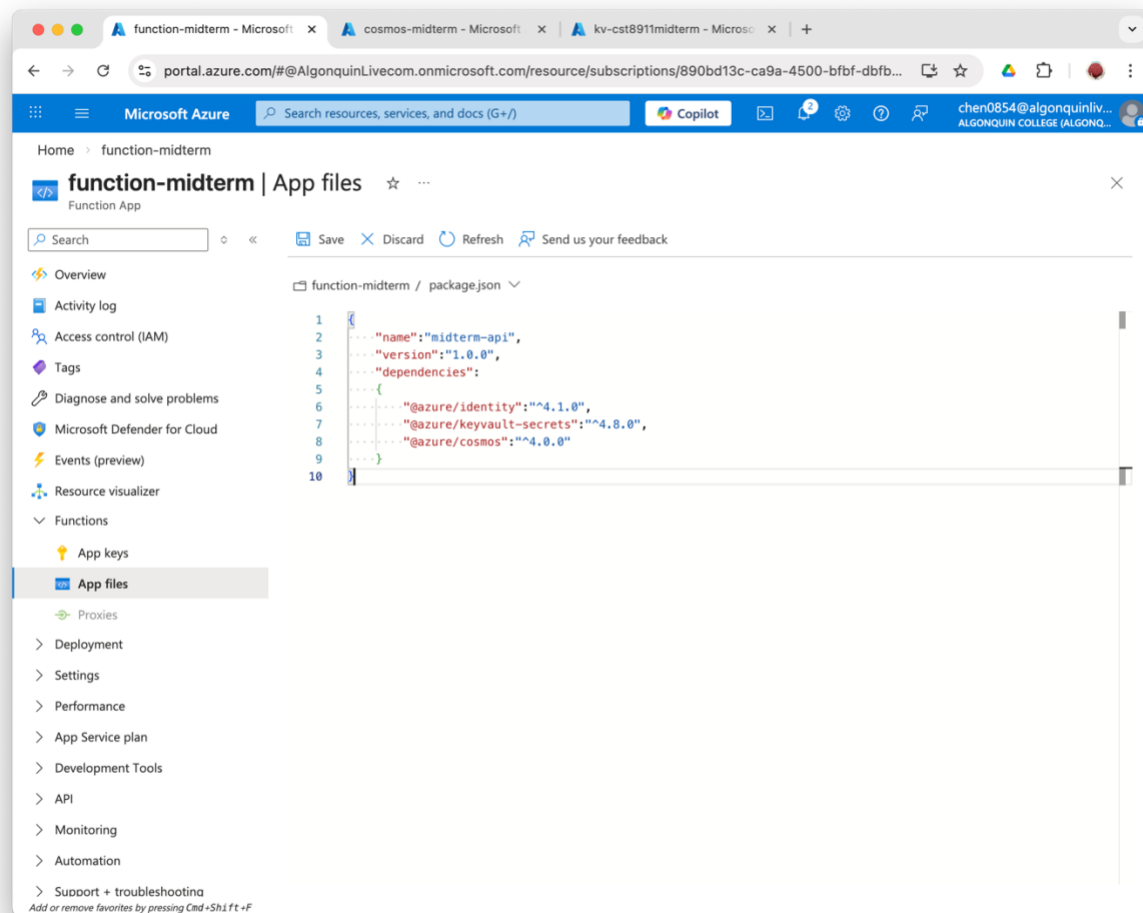
Name	Modified	Size
package.json	3/10/2026, 11:14:48 PM	
AddItem	3/10/2026, 10:49:58 PM	
DeleteItem	3/10/2026, 10:56:09 PM	
GetItems	3/10/2026, 10:49:22 PM	
host.json	3/10/2026, 10:06:39 PM	1 KB
package.json	3/10/2026, 11:14:10 PM	

Kudu Remote Execution Console
Type 'exit' then hit 'enter' to get a new powershell process.
Type 'cls' to clear the console

```
PS C:\home>
cd "C:\home\site"
PS C:\home\site>
cd "C:\home\site\wwwroot"
PS C:\home\site\wwwroot> touch package.json
touch package.json
Window title cannot be longer than 1023 characters.
PS C:\home\site\wwwroot> At line:1 char:1
+ touch package.json
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [], RuntimeException
+ FullyQualifiedErrorId : Argument

PS C:\home\site\wwwroot>
```

Go to App Files and add the dependencies code into **package.json**.



Test/Run the three functions to make sure it works.

GetItems (GET)

Test/Run - Microsoft Azure

cosmos-midterm - Microsoft

kv-cst8911midterm - Microso

+

portal.azure.com/#view/WebsitesExtension/FunctionTabMenuBlade/~/-codeTest/resourceId/%2Fsubscriptions%...

Microsoft Azure

Search resources, services, and docs (G+/)

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Home > function-midterm

GetItems | Code + Test

function-midterm

Code + Test

Integration

Function Keys

Invocations

Logs

Metrics

Save

Discard

Refresh

Test/Run

Get function URL

Disable

function-midterm / GetItems / index.js

```
1  const { DefaultAzureCredential } = require("@azure/identity");
2  const { SecretClient } = require("@azure/keyvault-secrets");
3  const { CosmosClient } = require("@azure/cosmos");
4
5  const COSMOS_ENDPOINT = process.env.COSMOS_ENDPOINT;
6  const COSMOS_KEY_SECRET_URI = process.env.COSMOS_KEY_SECRET_URI;
7  const COSMOS_DB_NAME = process.env.COSMOS_DB_NAME || "db-midterm";
8  const COSMOS_CONTAINER_NAME = process.env.COSMOS_CONTAINER_NAME || "ite
9
10 let _secretCache = { value: null, expiresAt: 0 };
11 function parseSecretUri(secretUri) {
12   const u = new URL(secretUri);
13   const parts = u.pathname.split("/").filter(Boolean);
14   if (parts[0] !== "secrets") throw new Error("Invalid COSMOS_KEY_SECRE
15   const name = parts[1];
16   const version = parts[2];
17   const vaultUrl = `${u.protocol}//${u.host}`;
18   return { vaultUrl, name, version };
19 }
20 async function getCosmosKey(context) {
21   const now = Date.now();
```

Logs

App Insights Logs

Connected! You are now viewing logs of Function runs in the current Code

'Logs' from the Function menu.

2026-03-11T03:29:08 [Information] Cosmos key pulled from Key Vault.

2026-03-11T03:29:08 [Information] Executed 'Functions.GetItems' (Su

2026-03-11T03:29:08 [Information] ("requestId":"c008d96d-e793-451b-

c4f29a81ca07","language":"node","sku":"Dynamic","dispatchDuration":34.3

Test/Run

Input

Output

Provide parameters to test the HTTP request. Results can be found in the Output tab.

HTTP method *

GET

Key *

default (Function key)

Query parameters

Name	Value	

Headers

Name	Value	

Run

Close

The screenshot displays the Microsoft Azure portal interface for testing a function. The main window is titled 'Test/Run' and shows the 'GetItems' function. The left pane contains the source code for 'index.js', which uses the Azure SDKs for Cosmos DB and Key Vault. The right pane shows the 'Test/Run' output, indicating a successful HTTP response with a 200 OK status. The logs pane at the bottom shows the function's execution details, including the Cosmos key being pulled from Key Vault.

```
1 const { DefaultAzureCredential } = require("@azure/identity");
2 const { SecretClient } = require("@azure/keyvault-secrets");
3 const { CosmosClient } = require("@azure/cosmos");
4
5 const COSMOS_ENDPOINT = process.env.COSMOS_ENDPOINT;
6 const COSMOS_KEY_SECRET_URI = process.env.COSMOS_KEY_SECRET_URI;
7 const COSMOS_DB_NAME = process.env.COSMOS_DB_NAME || "db-midterm";
8 const COSMOS_CONTAINER_NAME = process.env.COSMOS_CONTAINER_NAME || "items";
9
10 let _secretCache = { value: null, expiresAt: 0 };
11 function parseSecretUri(secretUri) {
12   const u = new URL(secretUri);
13   const parts = u.pathname.split("/").filter(Boolean);
14   if (parts[0] !== "secrets") throw new Error("Invalid COSMOS_KEY_SECRET_URI");
15   const name = parts[1];
16   const version = parts[2];
17   const vaultUrl = `${u.protocol}//${u.host}`;
18   return { vaultUrl, name, version };
19 }
20 async function getCosmosKey(context) {
21   const now = Date.now();
```

Test/Run

Input Output

HTTP response code 200 OK

HTTP response content

Run Close

Connected! You are now viewing logs of Function runs in the current Code editor. Click 'Logs' from the Function menu.

```
2026-03-11T03:29:08 [Information] Cosmos key pulled from Key Vault.
2026-03-11T03:29:08 [Information] Executed 'Functions.GetItems' (Su
2026-03-11T03:29:08 [Information] ("requestId":"c008d96d-e793-451b-
c4f29a81ca07","language":"node","sku":"Dynamic","dispatchDuration":34.3
```

AddItem (POST)

Test/Run - Microsoft Azure

cosmos-midterm - Microsoft

kv-cst8911midterm - Microso

+

portal.azure.com/#view/WebsitesExtension/FunctionTabMenuBlade/~/-codeTest/resourceId/%2Fsubscriptions%...

Microsoft Azure

Search resources, services, and docs (G+)

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Home > function-midterm

AddItem | Code + Test ...

function-midterm

Code + Test Integration Function Keys Invocations Logs Metrics

Save Discard Refresh Test/Run Get function URL Disable

function-midterm / AddItem / index.js

```
1 const { DefaultAzureCredential } = require("@azure/identity");
2 const { SecretClient } = require("@azure/keyvault-secrets");
3 const { CosmosClient } = require("@azure/cosmos");
4
5 const COSMOS_ENDPOINT = process.env.COSMOS_ENDPOINT;
6 const COSMOS_KEY_SECRET_URI = process.env.COSMOS_KEY_SECRET_URI;
7 const COSMOS_DB_NAME = process.env.COSMOS_DB_NAME || "db-midterm";
8 const COSMOS_CONTAINER_NAME = process.env.COSMOS_CONTAINER_NAME || "ite
9
10 let _secretCache = { value: null, expiresAt: 0 };
11 function parseSecretUri(secretUri) {
12   const u = new URL(secretUri);
13   const parts = u.pathname.split("/").filter(Boolean);
14   if (parts[0] !== "secrets") throw new Error("Invalid COSMOS_KEY_SECRET_URI");
15   const name = parts[1];
16   const version = parts[2];
17   const vaultUrl = `${u.protocol}//${u.host}`;
18   return { vaultUrl, name, version };
19 }
20 async function getCosmosKey(context) {
21   const now = Date.now();
```

Logs App Insights Logs

Connected! You are now viewing logs of Function runs in the current Code Editor. Logs from the Function menu.

Test/Run

Input Output

Provide parameters to test the HTTP request. Results can be found in the Output tab.

HTTP method * POST

Key * default (Function key)

Query parameters

Name	Value

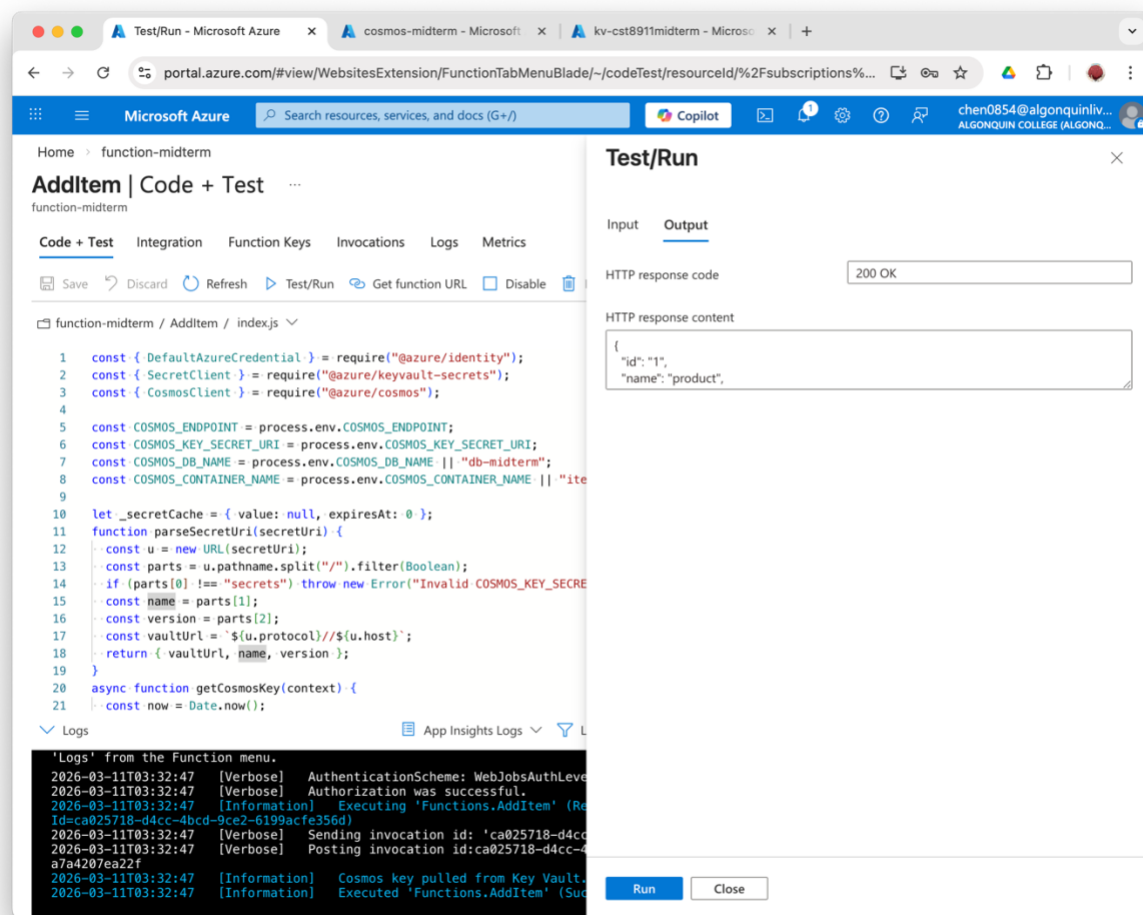
Headers

Name	Value

Body

```
1 {
2   "id": "1",
3   "name": "product",
4   "price": 9.99
5 }
```

Run Close



DeleteItem (DELETE)

Edit the Trigger of **DeleteItem** first to have the correct HTTP method and route, and then Test/Run.

portal.azure.com/#view/WebsitesExtension/FunctionTabMenuBlade/~integration/resourceId/%2Fsubscriptions...

Microsoft Azure

Search resources, services, and docs (G+)

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Home > function-midterm > Deleteltem

Deleteltem | Integration

function-midterm

Code + Test Integration Function Keys Invocations Logs Metrics

Refresh Send us your feedback

Trigger and inputs

Trigger

HTTP (req)

Inputs

No inputs defined

Add input

Function

Deleteltem

Edit Trigger

Binding Type * ⓘ

HTTP

HTTP details

Request parameter name * ⓘ

req

Route template ⓘ

items/{id}

Authorization level * ⓘ

Function

Selected HTTP methods ⓘ

GET, POST, DELETE

Save

Cancel

Delete

Test/Run - Microsoft Azure

cosmos-midterm - Microsoft

kv-cst8911midterm - Microso

+

portal.azure.com/#view/WebsitesExtension/FunctionTabMenuBlade/~/-codeTest/resourceId/%2Fsubscriptions%...

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Home > function-midterm

Deleteltem | Code + Test ...
function-midterm

Code + Test Integration Function Keys Invocations Logs Metrics

Save Discard Refresh Test/Run Get function URL Disable

function-midterm / Deleteltem / index.js

```
1 const { DefaultAzureCredential } = require("@azure/identity");
2 const { SecretClient } = require("@azure/keyvault-secrets");
3 const { CosmosClient } = require("@azure/cosmos");
4
5 const COSMOS_ENDPOINT = process.env.COSMOS_ENDPOINT;
6 const COSMOS_KEY_SECRET_URI = process.env.COSMOS_KEY_SECRET_URI;
7 const COSMOS_DB_NAME = process.env.COSMOS_DB_NAME || "db-midterm";
8 const COSMOS_CONTAINER_NAME = process.env.COSMOS_CONTAINER_NAME || "ite
9
10 let _secretCache = { value: null, expiresAt: 0 };
11 function parseSecretUri(secretUri) {
12   const u = new URL(secretUri);
13   const parts = u.pathname.split("/").filter(Boolean);
14   if (parts[0] !== "secrets") throw new Error("Invalid COSMOS_KEY_SECRE
15   const name = parts[1];
16   const version = parts[2];
17   const vaultUrl = `${u.protocol}//${u.host}`;
18   return { vaultUrl, name, version };
19 }
20 async function getCosmosKey(context) {
21   const now = Date.now();
```

Logs App Insights Logs

Connected! You are now viewing logs of Function runs in the current Code 'Logs' from the Function menu.

2026-03-11T03:44:05 [Information] Cosmos key pulled from Key Vault.

2026-03-11T03:44:05 [Information] Deleteltem attempting: id="1" pk

2026-03-11T03:44:05 [Information] Executed 'Functions.Deleteltem'

Test/Run

Input Output

Provide parameters to test the HTTP request. Results can be found in the Output tab.

HTTP method *

DELETE

Key *

default (Function key)

Query parameters

Name	Value
id	1

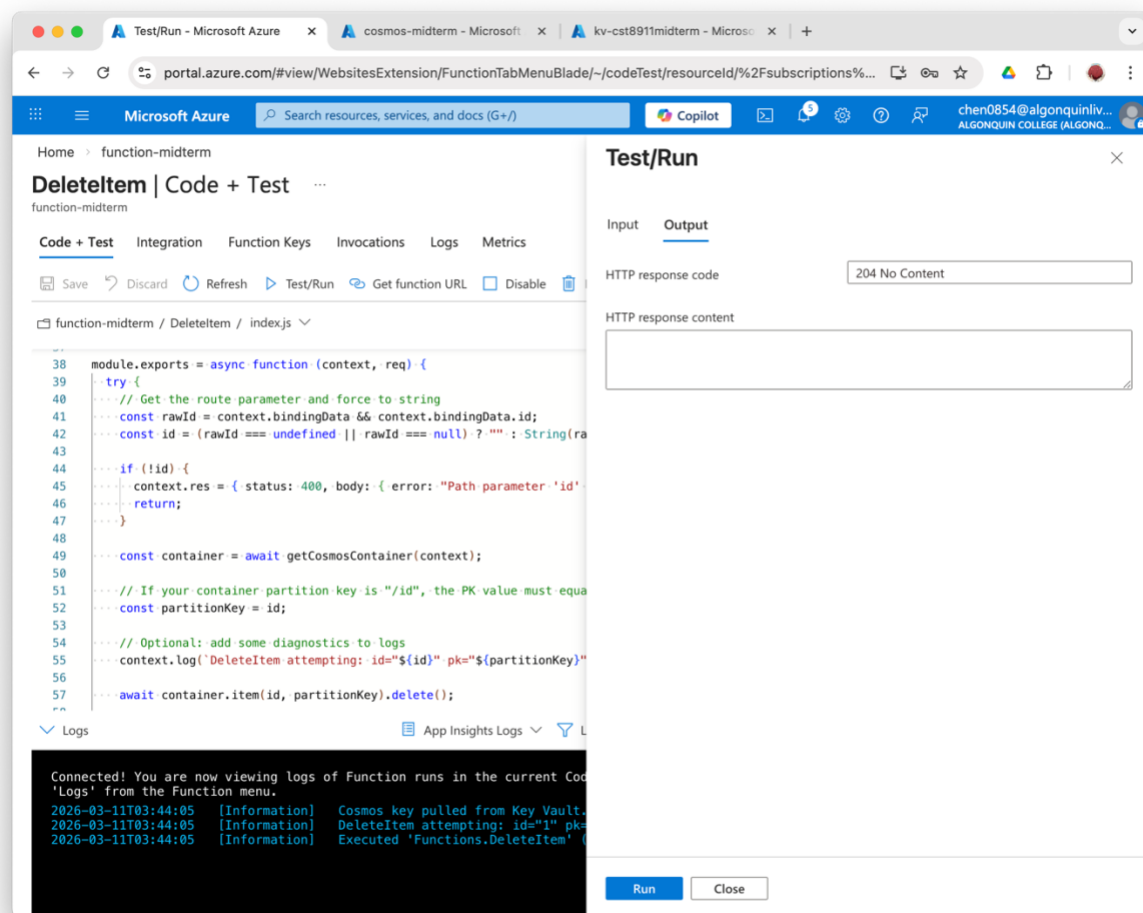
Headers

Name	Value

Body

1

Run Close

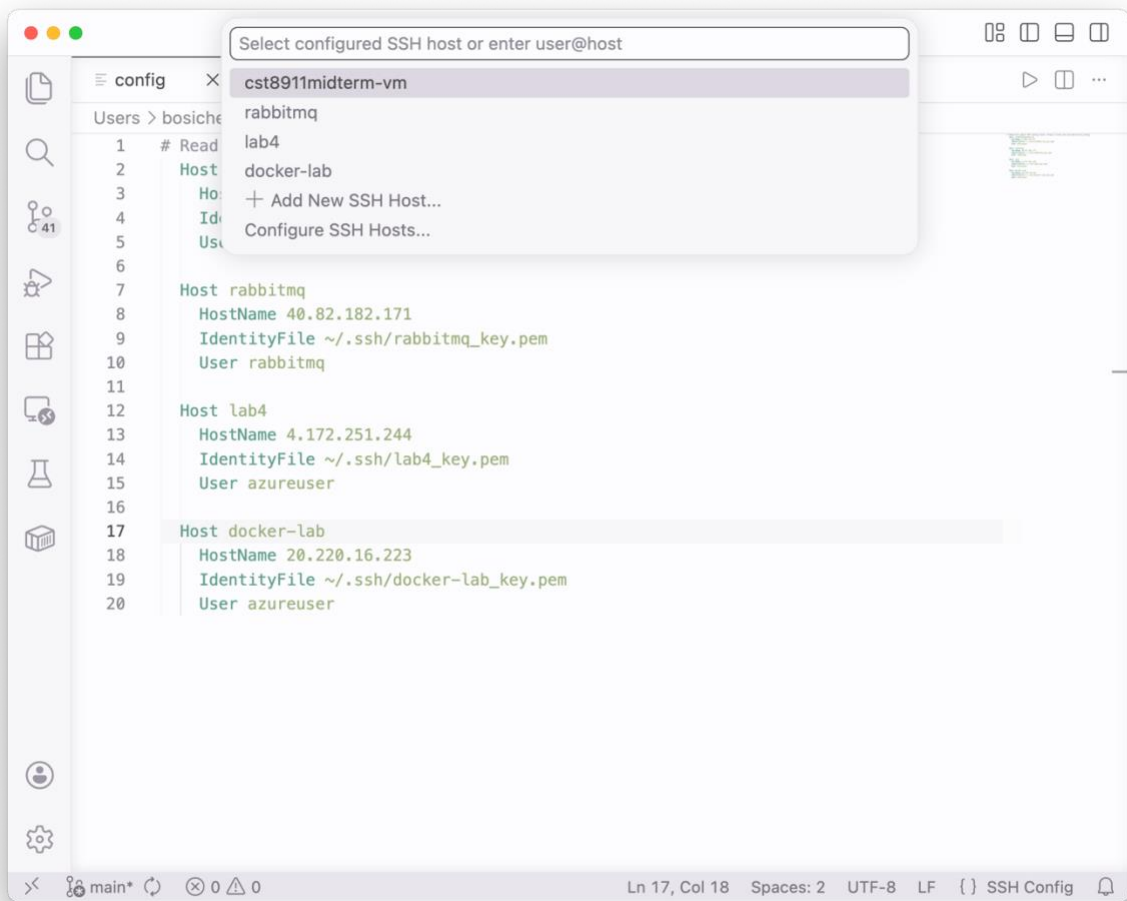


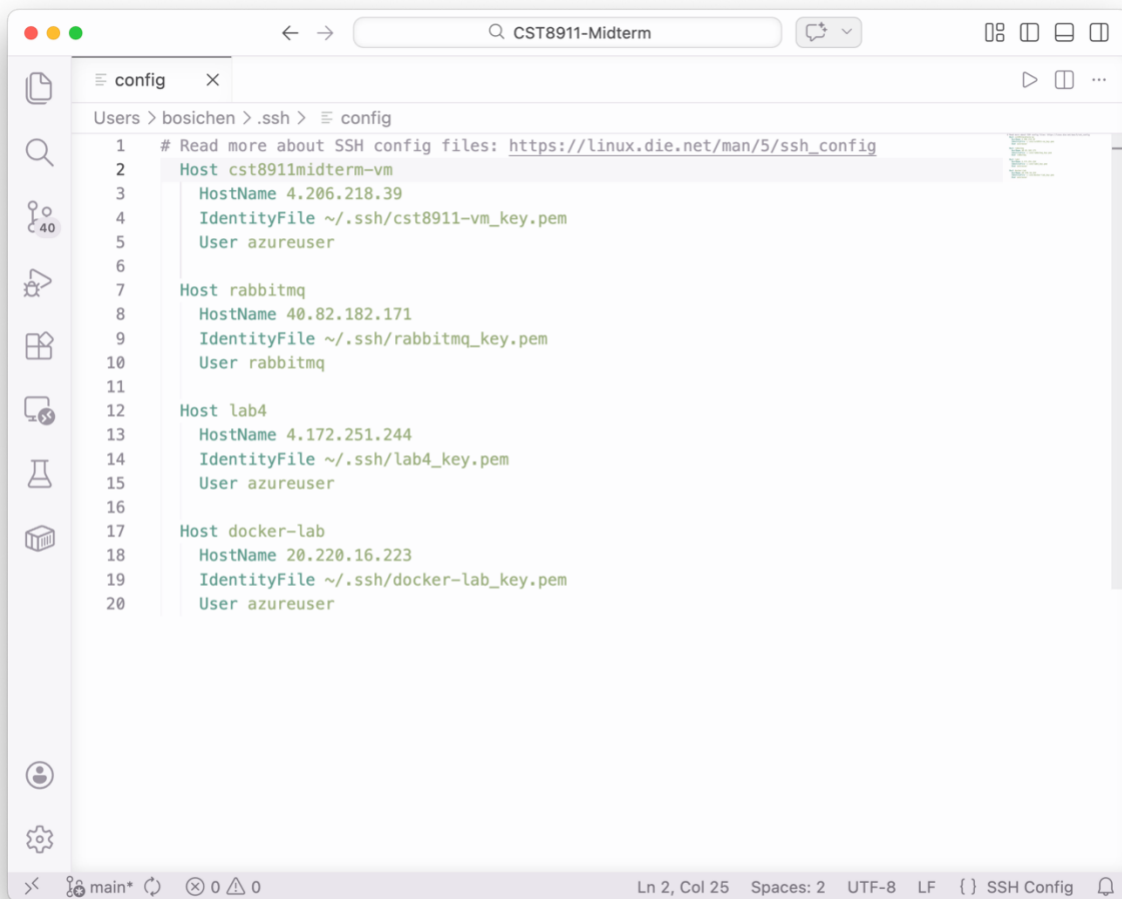
10. Test the functions from virtual machines

Use VSCode to SSH into the VM created in the previous steps.

Command Palette --> Connect to Host --> Add New SSH Host --> **ssh -i ~/.ssh/cst8911-vm_key.pem azureuser@4.206.218.39** --> save it to the ssh config file and change the **Host** to **cst8911midterm-vm**

Then the new Host will show up when opening the Command Palette again. Select Connect to Host --> click to connect.

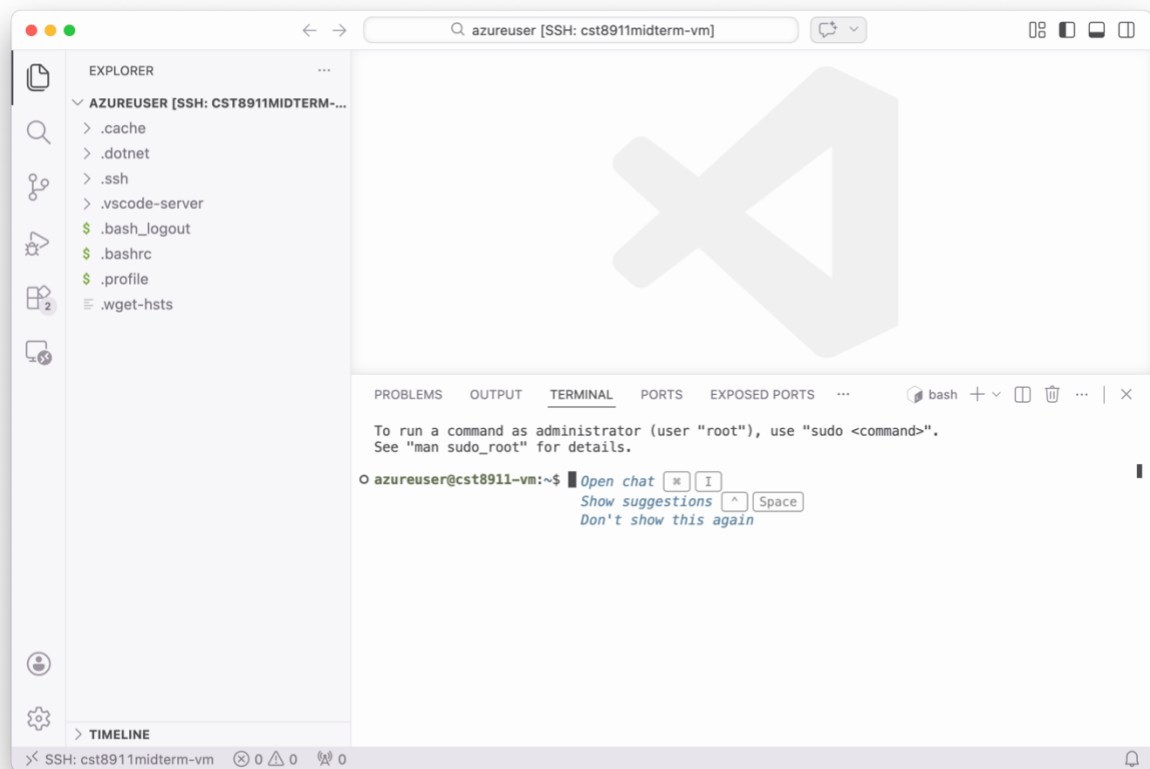




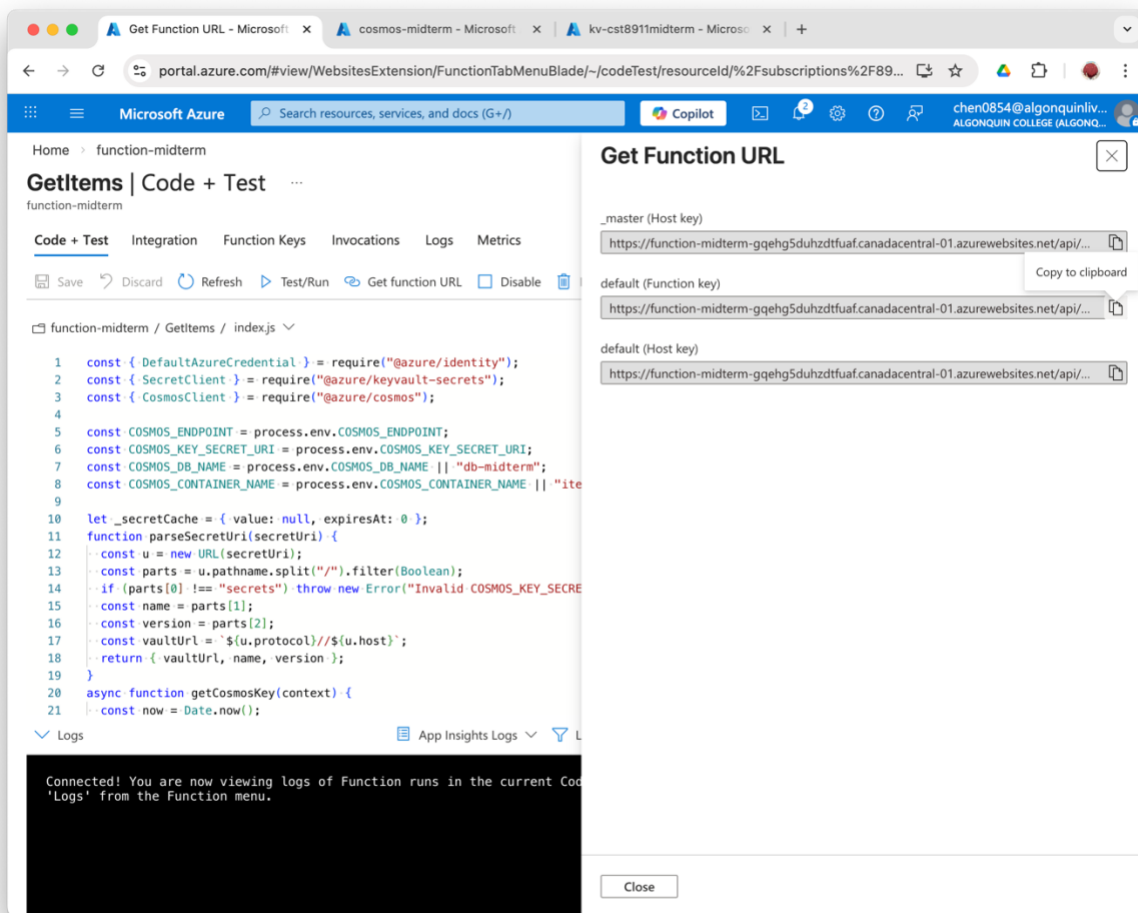
The image shows a code editor window titled "CST8911-Midterm". The editor is displaying an SSH configuration file named "config". The file content is as follows:

```
1 # Read more about SSH config files: https://linux.die.net/man/5/ssh\_config
2 Host cst8911midterm-vm
3   HostName 4.206.218.39
4   IdentityFile ~/.ssh/cst8911-vm_key.pem
5   User azureuser
6
7 Host rabbitmq
8   HostName 40.82.182.171
9   IdentityFile ~/.ssh/rabbitmq_key.pem
10  User rabbitmq
11
12 Host lab4
13   HostName 4.172.251.244
14   IdentityFile ~/.ssh/lab4_key.pem
15   User azureuser
16
17 Host docker-lab
18   HostName 20.220.16.223
19   IdentityFile ~/.ssh/docker-lab_key.pem
20   User azureuser
```

The editor interface includes a sidebar with various icons (file explorer, search, source control, etc.), a top bar with navigation and search controls, and a bottom status bar showing the current file path, line and column numbers, and encoding information.

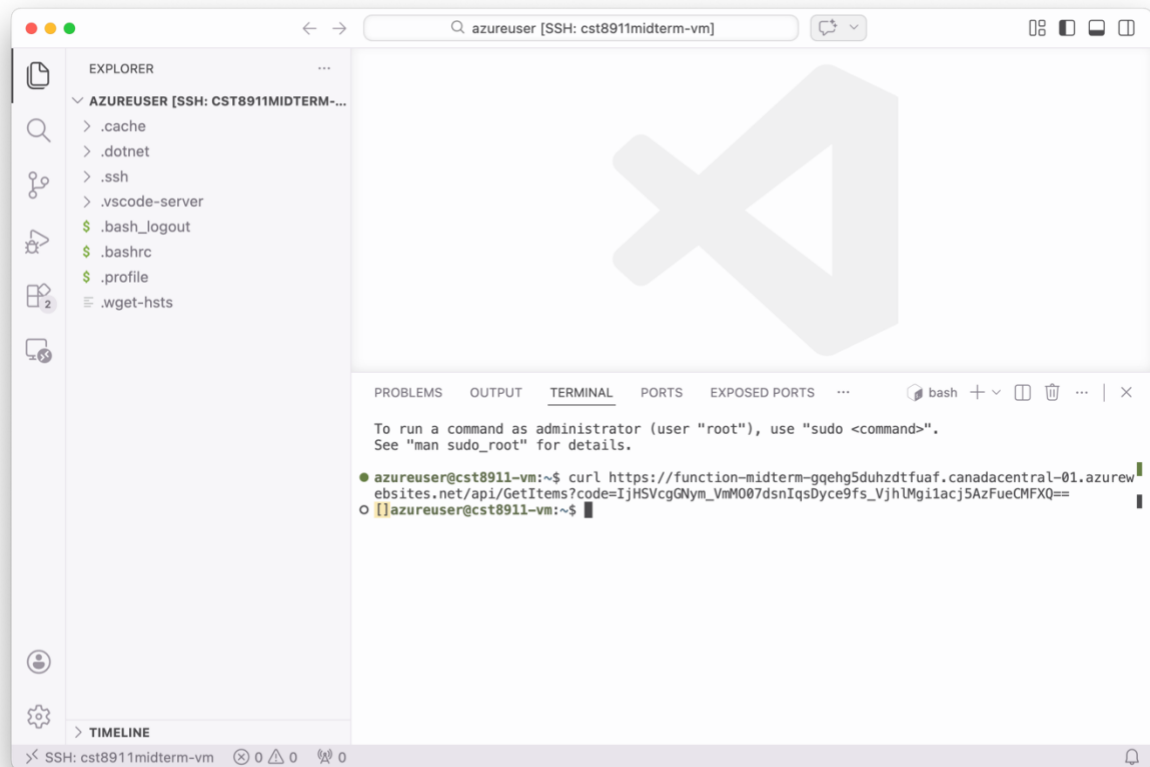


Copy the function url from the Code + Test page



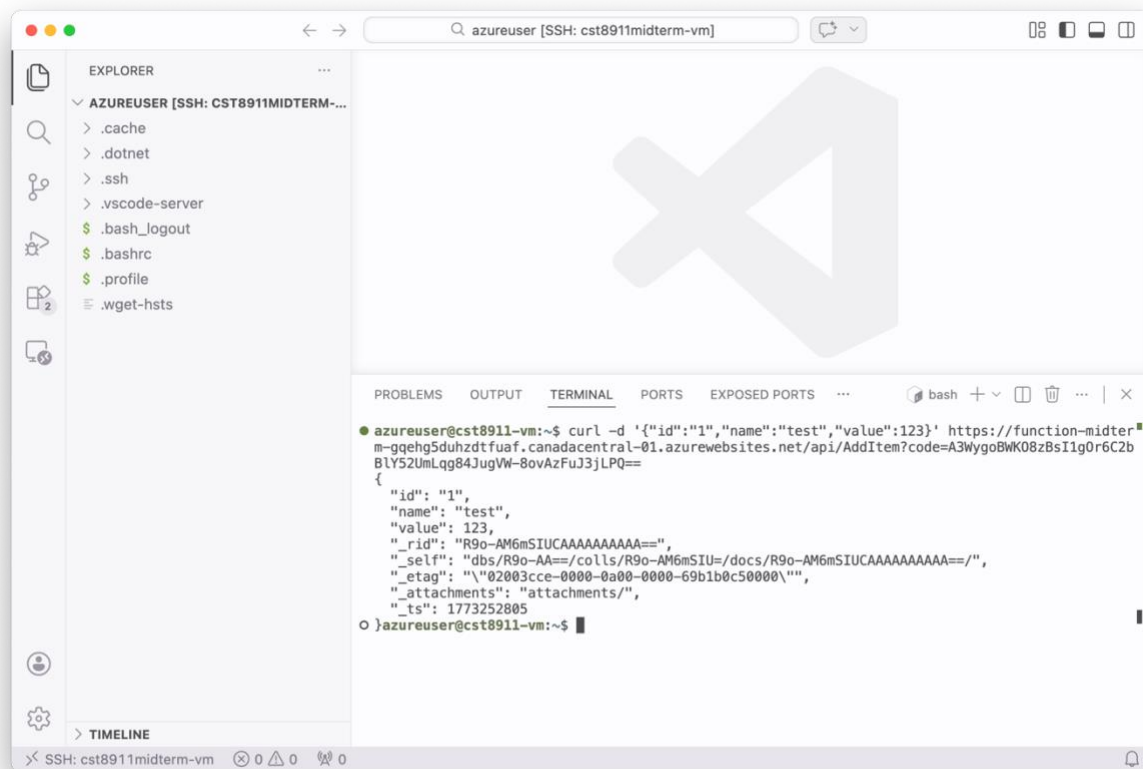
curl to test the **GetItems** function:

Returned [] initially since the container is empty.



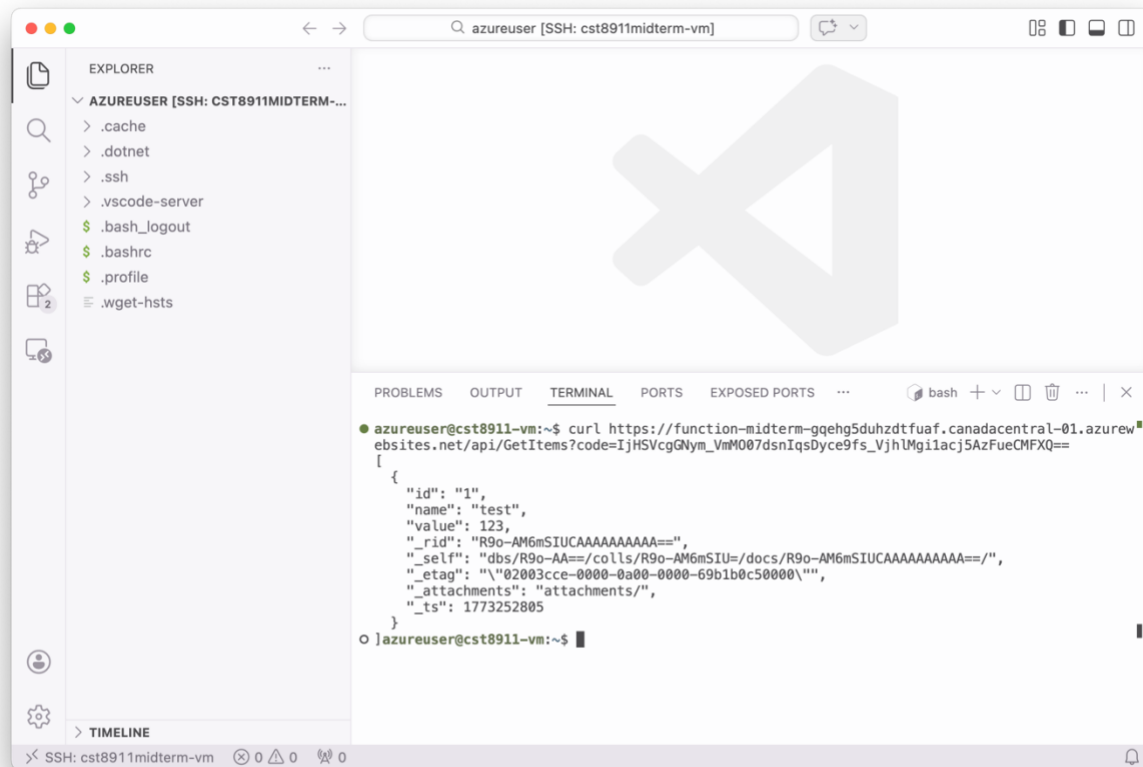
curl to test the **AddItem** function:

Returned the inserted JSON item



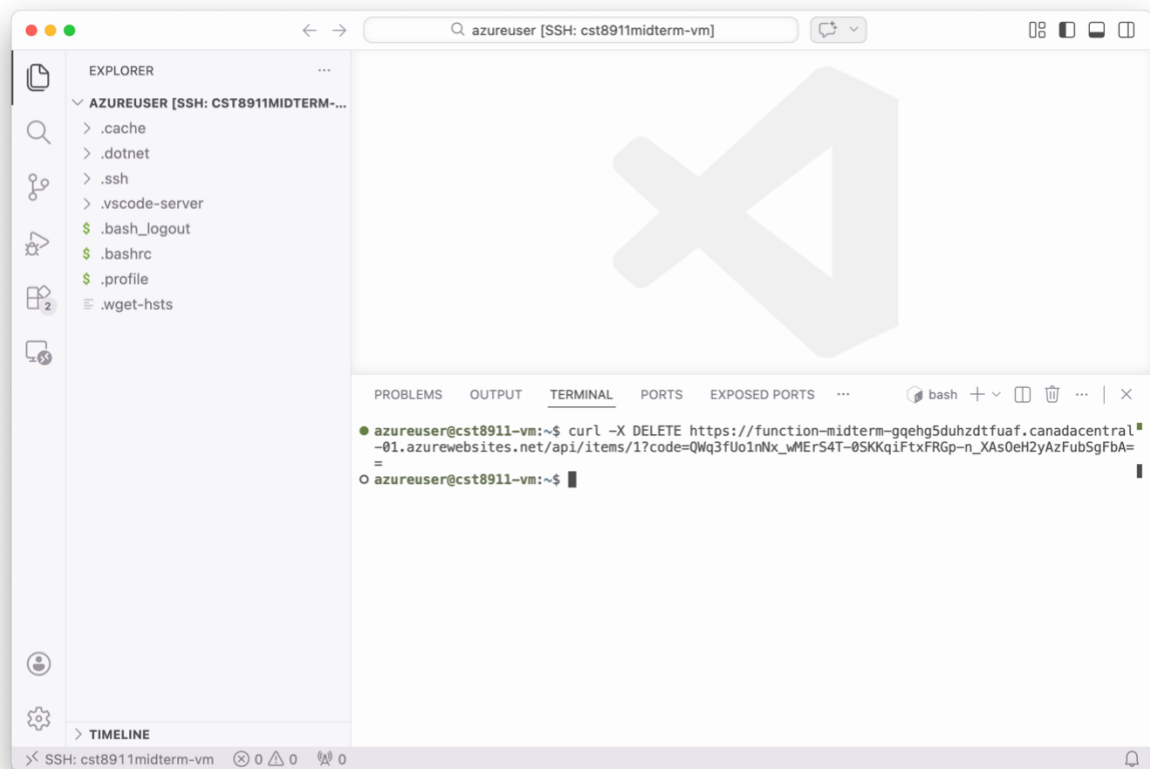
Call the **GetItems** function again to see if the new item is successfully added:

Returned the container with the newly added item.



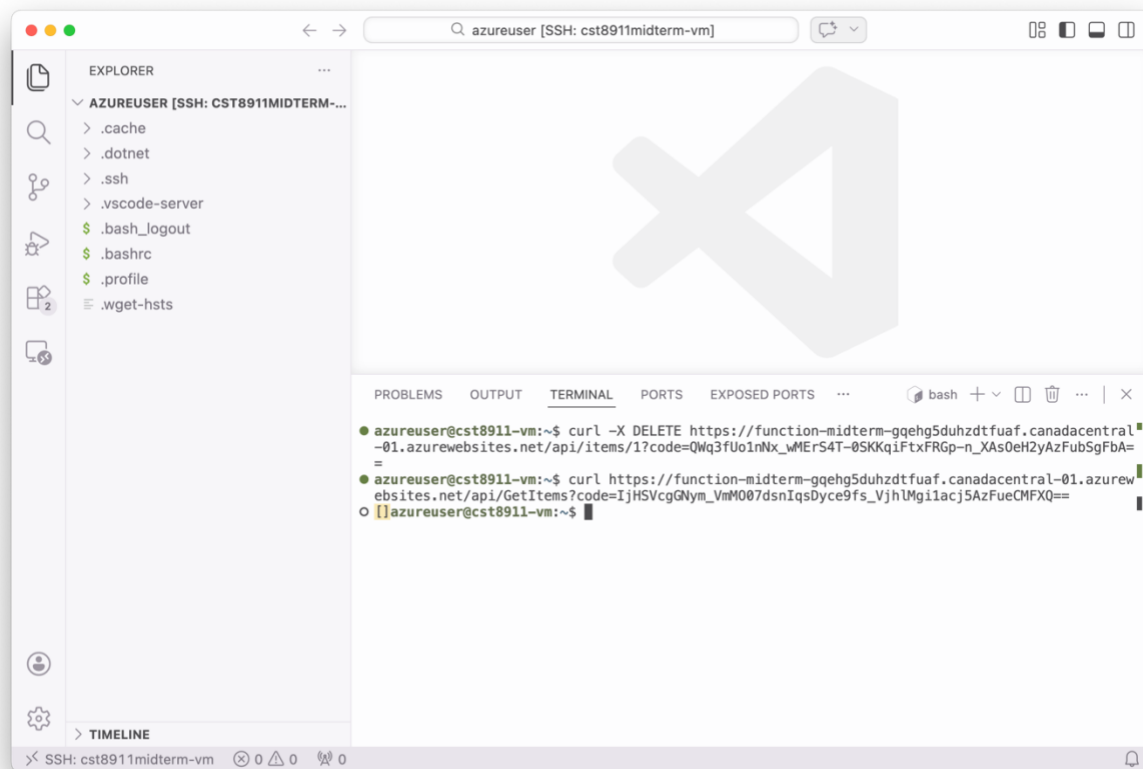
curl to test the **DeleteItem** function:

Should return (204) no content



Call the **GetItems** function again to see if the new item is successfully deleted:

Returned [] again



11. Delete all the resources

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

midterm

14 resources being deleted across 1 subscriptions, 2 resource groups
All dependent resources, including hidden types, are shown

Name	Resource type	Subscription	Resource Group
ASP-midterm-836c	App Service plan	Azure for Students	midterm
cosmos-midterm	Azure Cosmos DB account	Azure for Students	midterm
cst8911-vm	Virtual machine	Azure for Students	midterm
cst8911-vm-ip-096463a7	Public IP address	Azure for Students	midterm
cst8911-vm-nsg	Network security group	Azure for Students	midterm
cst8911-vm-vnet	Virtual network	Azure for Students	midterm
cst8911-vm806-096463a7	Network interface	Azure for Students	midterm
cst8911-vm_disk1_9d043a2857b64a69b1fdbfd46	Disk	Azure for Students	MIDTERM
cst8911-vm_key	SSH key	Azure for Students	midterm
function-midterm	Application Insights	Azure for Students	midterm

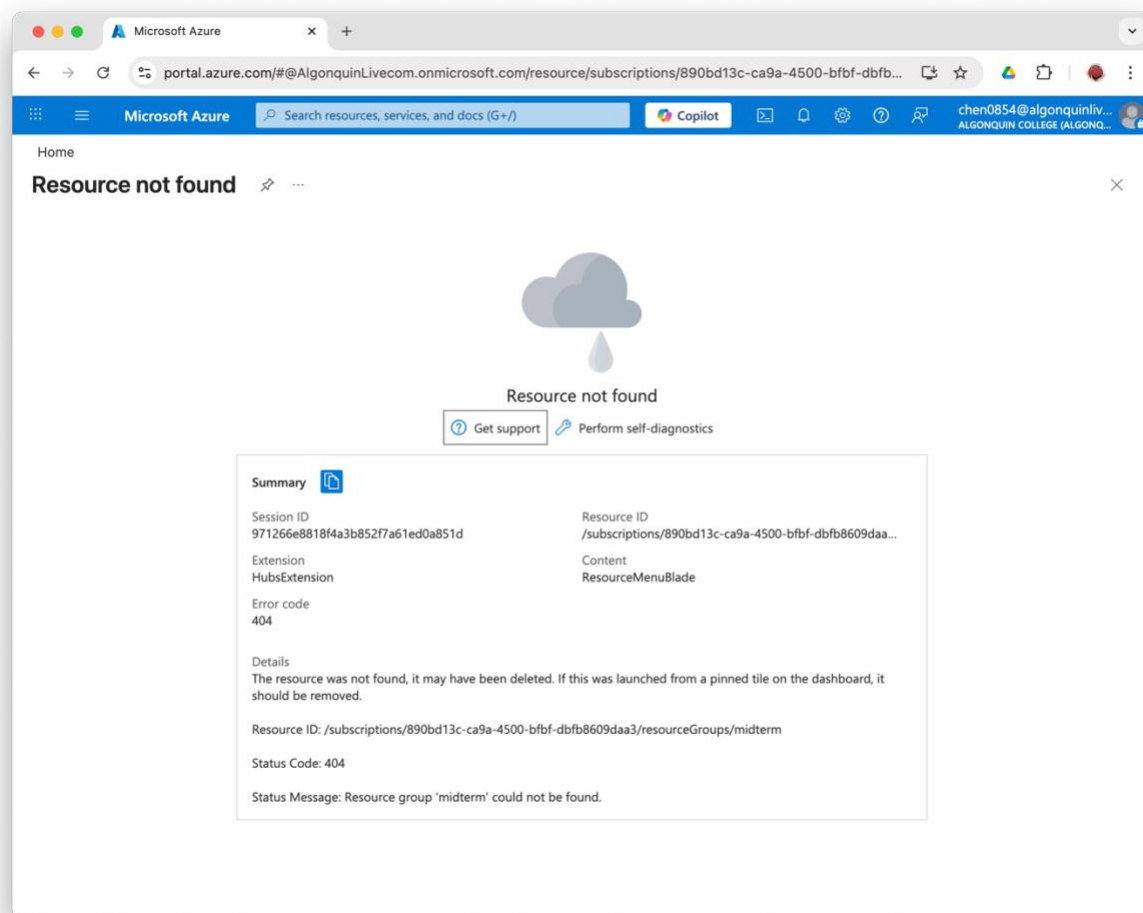
☒ Apply force delete for selected Virtual machines and Virtual machine scale sets ⓘ

Enter resource group name to confirm deletion *

midterm

Delete **Cancel**

Add or remove favorites by pressing Cmd+Shift+F



Conclusion

This project shows how a secured, scalable serverless RESTful API function app is created in Azure in a cost-effective way. The Key Vault can help storing secret values securely and the Managed Identity can reduce the complexity of security management.